

“KNOW BEFORE. ACT BEFORE.”

How can we turn **climate information** into ***food security resilience?***

Team : **JJ Vision**

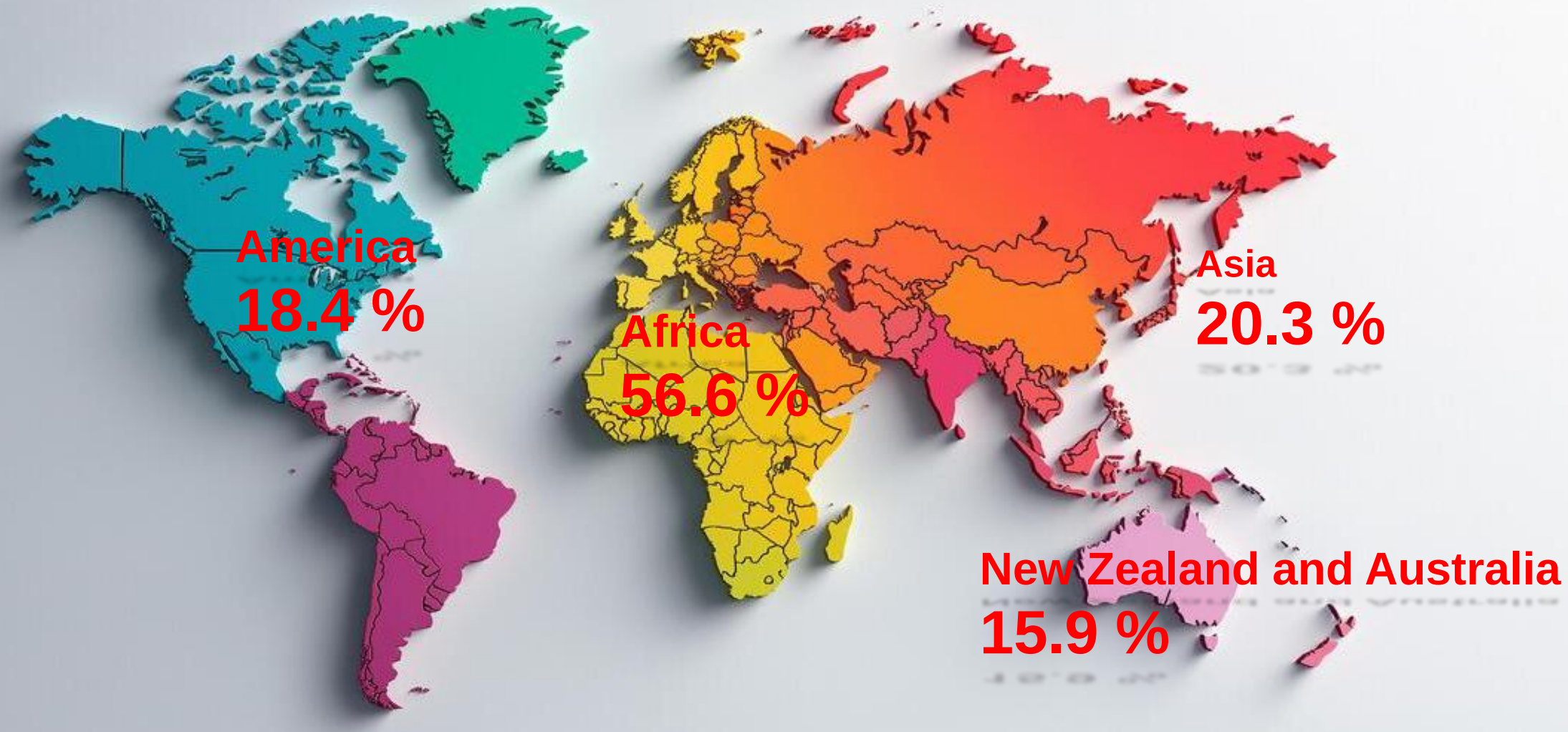
Members :

Tan Jia Wei

Wong Jun Xiang

2.5 billion – 1 in 4 people went without reliable access to enough food in 2025

The State of Food Security and Nutrition in the World (2026)



Prevalence of moderate or severe food insecurity in the total population (2025)

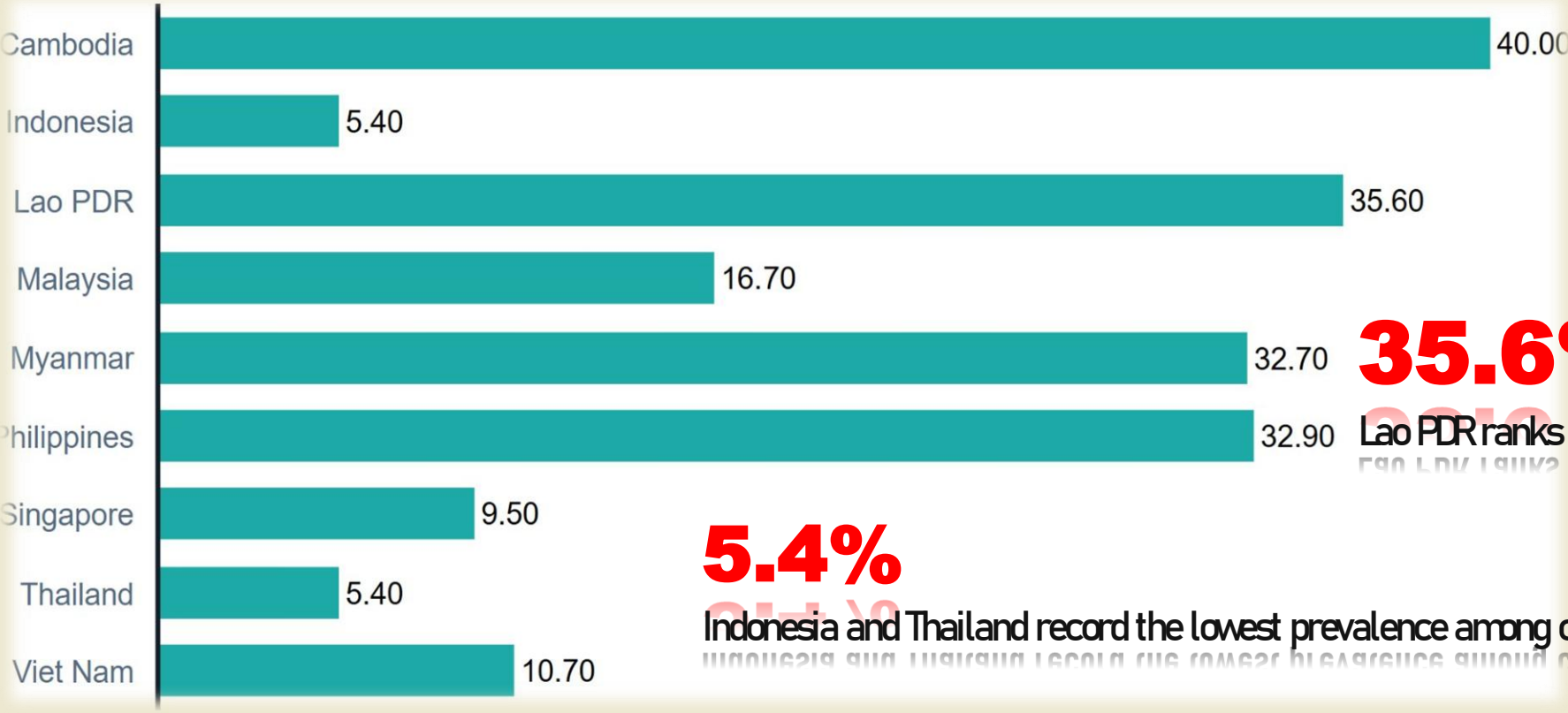
Source : FAOSTAT- Suite of Food Security Indicators



12.8% 87.6 million people

ASEAN population experiencing moderate or severe **food insecurity** in **2023-2025** (FAO)

Prevalence of Moderate or Severe Food Insecurity in ASEAN Countries (%) (2022–2024)



40.0%

Cambodia has the highest prevalence of moderate or severe food insecurity.

35.6%

Lao PDR ranks second among the ASEAN countries shown.

5.4%

Indonesia and Thailand record the lowest prevalence among countries with available data.

Food insecurity remains a significant challenge **across ASEAN, with substantial differences between countries.**

(FAO, IFAD, UNICEF, WFP & WHO (2025), The State of Food Security and Nutrition in the World 2025)



THE STORIES BEHIND *FOOD SHORTAGE*

Exploring the **key triggers on every Asean members , underlying issues, and impacts** of food shortages on Malaysia.

Food shortages don't happen overnight....

Why Does Food Insecurity Still Remain Crisis In Malaysia ?

Baseline Vulnerabilities 1 : <u>Production</u>

- Limited domestic production leaves less buffer to absorb supply shocks, making food shortages more likely.

Malaysia's **paddy production fell** from **2.18 million** tonnes in 2023 **to 2.06 million** tonnes in 2024 — **a 5.3% decline**. (DOSM,2025)

Insufficient Domestics Production : **Rice** fell to **52.9% in 2024**; beef **16.8%**, mutton **8.5%**, mango **20.6%** （SSR）

Pest or disease losses : Brown planthopper can cause **up to 60%** rice losses (MARDI – Pest early warning technology)

Rice bacterial disease : BPB affected **at least 12,000 ha**; yield losses up to **50%** reported (MARDI – Rice research)

Food shortages don't happen overnight....

Why Does Food Insecurity Still Remain Crisis In Malaysia ?

Baseline Vulnerabilities 2 : Supply-chain

- **production loss** from **disruption after production**.
- Weak storage and distribution cause food losses after production, reducing the supply that actually reaches consumers.

Lack of Cold-Storage Facilities : DOSM identifies **lack of mobile cold storage, low-temperature warehouses and temporary storage centres** as contributors to higher post-harvest losses.(DOSM — Household Food Waste / Food System Report)

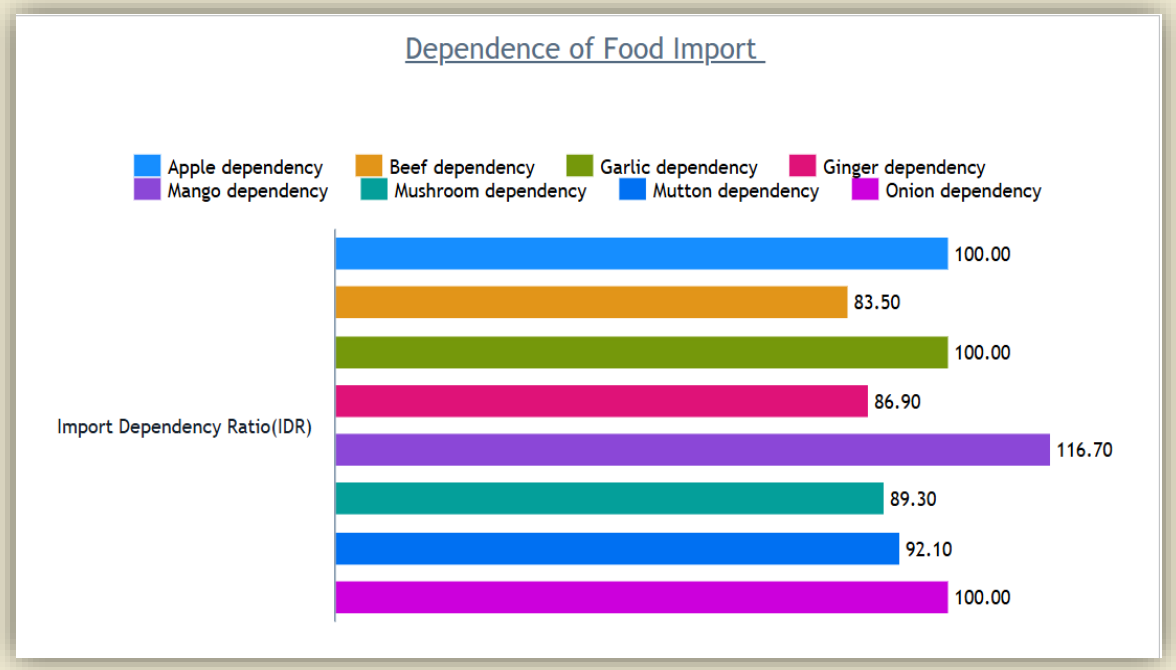


Food shortages don't happen overnight....

Why Does Food Insecurity Still Remain Crisis In Malaysia ?

Baseline Vulnerabilities 3 : Import dependence

Overall food import in 2024 : **RM93.8 billion food imports** in 2024
(DOSM — SUA 2024)
Increasing Import : **Food imports** increased **19.1%** from 2023
(DOSM — SUA 2024)



Climate Change : The Multiplier

Many factors drive food shortages. Why climate change?
Malaysia's food system has existing vulnerabilities — from production and supply dependence to water and cost pressures. But these vulnerabilities do not always create a crisis on their own. Climate change is different because it can activate several of them at the same time. A flood, drought or extreme heat event can disrupt production, water, transport and supply simultaneously.



FAO forecasts Malaysia's **2026 paddy production at 2.1 million tonnes**, which is **4.5% below the five-year average**, primarily because of **flooding problems and dry-weather conditions**

- **Rice is affected by erratic rainfall, extreme heat and frequent flooding**
- **Rubber is sensitive to drought and precipitation changes.**
- **Palm oil also affected by erratic rainfall and extreme temperatures, disrupting flowering and oil quality**

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How climate change disrupt food supply chain

Climate change

Etc.Extreme rainfall / floods

↓

- **RM303.4 million** damage to public assets and infrastructure
- **RM54.1 million** damage to business premises

DOSM,2024

↓

Transportation / distribution disrupted

↓

Food availability & prices affected

Climate Change : The Multiplier

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FAO 2026:

- Malaysia's **2026 paddy production** is forecast at **2.1 million tonnes**, 4.5% below the five-year average, with flooding and dry weather contributing to reduced planting and yields.
- At the same time, Malaysia's **2026/27 cereal import requirement** is forecast at **7.9 million tonnes**, around **10% above the average**.

Climate change makes water availability more variable and difficult to manage, high temperatures and low rainfall as exacerbating drought, while intense rainfall increases flood risk (Malaysia Climate Knowledge Portal ,World Bank)



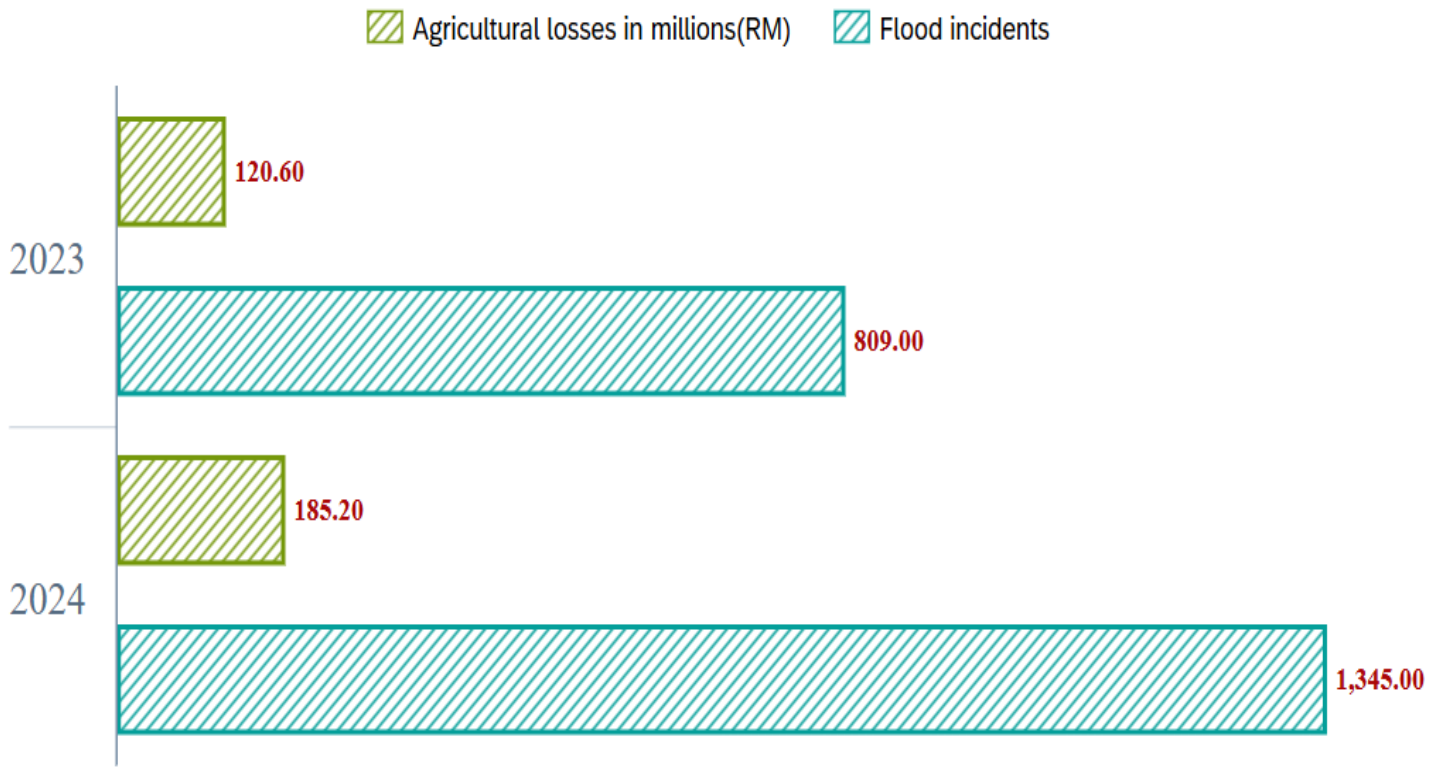
Year 2024 -1345 flood incidents, flood-related agricultural losses – **RM 185.2 million**

Rising temperatures, droughts, floods and changing rainfall as threats to the production of rice and other crops (Malaysia Economic Monitor ,World Bank)

Year 2025 – RM 52.6 million losses ,all states were affected by **flood**...

"Stronger storms, longer dry spells and unpredictable monsoons are putting **food supply chains under pressure" – World Bank**

Agricultural losses in millions(RM) , Flood incidents per Year



From 2023 → 2024

809 floods → RM120.6M agricultural loss

↓

1,345 floods → RM185.2M agricultural loss

Flood incidents: +66.3%
Agricultural losses: +53.5%

(DOSM — Special Report on Impact of Floods in Malaysia, 2024)

The Systemic Root Causes : Why does one climate shock amplify multiple vulnerabilities?

Fragmented information

Insufficient information connected, informations exists but scattered across different agencies and system.

Malaysia lacks a unified strategy integrating hazard data ,vulnerability assessments and community preparedness into one cohesive multi-hazard early-warning system .(OECD,2025)





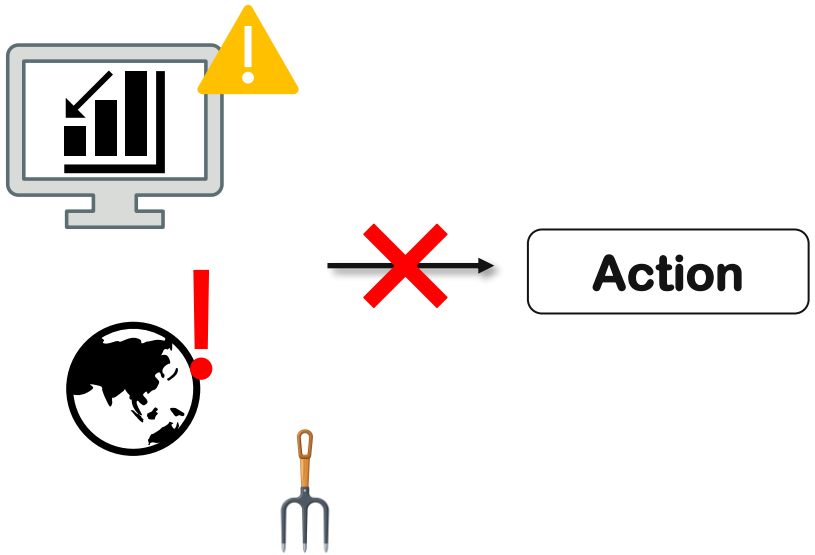
The Systemic Root Causes : Why does one climate shock amplify multiple vulnerabilities?

Fragmented information

Malaysia have prediction to climate-related hazard but frustrated in translating it into timely action

OECD identifies challenges in **scaling and institutionalising impact-based forecasting**, including differences in local conditions, institutional coordination, capacity and infrastructure. Malaysia's early-warning systems also face challenges in turning hazard information into **actionable, coordinated responses**.(OECD,2025)

Prediction–action gap



The Systemic Root Causes : Why does one climate shock amplify multiple vulnerabilities?

Fragmented information

Prediction–action gap

Insufficient Resilience

<u>Climate event exposes vulnerabilities, weak recovery and adaptation allow those weaknesses to persist, causing greater losses when the next shock occurs.</u>

-World Bank had stated that Malaysia need for greater **productivity, efficiency and resilience** in agrofood sector, specifically climate-resilient seeds, water-efficient irrigation, weather information, early-warning systems and crop insurance.(World Bank)



The Systemic Root Causes : Why does one climate shock amplify multiple vulnerabilities?

Fragmented information

Prediction–action gap

Insufficient Resilience

Information Gaps



<u>Missing or insufficient data such as localized crops conditions makes it difficult to identify risks early and accurately.</u>

40% of 143 critical data items were unavailable , only 18% of data were readily available.

(World Bank Document,2023)





SOLUTION :

FOODWISE

Target :

Turns climate information into actionable food-security intelligence

Identify climate risks, highlight potential shortages ,decision-supporter

fWise

FWise :A data-driven food security resilience platform

Fwise aim to connect fragmented data, support decision-making, and link government agencies with operational users so information can be translated into action.

- Integrated Data Hub
- IoT Climate Monitoring
- Collaborative Network
- Predictive & Action-Oriented





Two Main Users Of FWise Application



Data Providers

Government / Coordinating Agencies

Role :

- Monitor the current situation and receive CRIS alerts
- Review **AI-generated strategy options**
- **Authorize the overall response strategy**
- Monitor implementation across all stakeholders





Implementation Users

Registered Food / Supply-Chain Organizations

Role :

- Receive their assigned CRIS directive
- View the relevant forecast
- View the **already-authorized strategy**
- See exactly what their organization needs to do
- Update implementation progress
- Upload evidence

Introducing IOT Monitoring System : **Automatic Weather Station (AWS)**



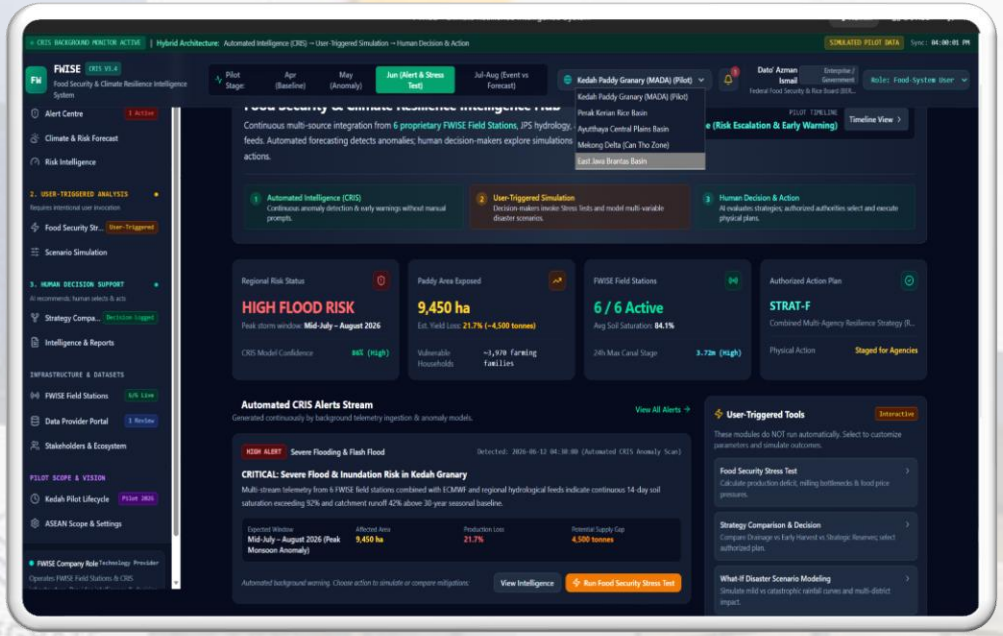
WMO/NOAA systems commonly combine sensors for temperature, humidity, precipitation, wind, radiation, pressure, soil conditions, plus a data logger and communications system

(World Meteorological Organization , 2025)

fWise

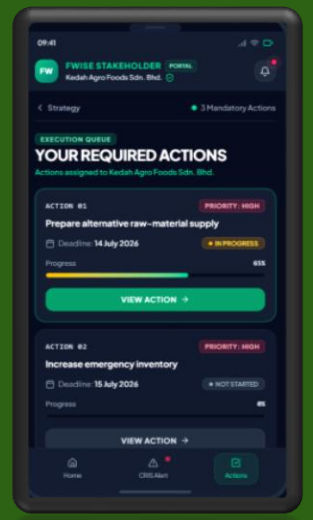
Food Security & Climate Resilience Intelligence System (CRIS)

Main FWise CRIS platform



- Central intelligence platform for data providers and decision-makers
- Integrate information and data from multiple sources
- Continue analyses conditions and identify emerging risks
- AI –assisted scenario simulation and strategics generation
- Action report and tasks distribution

FWise Stakeholder mobile app.



- The operational platform connects stakeholders.
- delivers risk alerts and recommended actions directly to relevant stakeholders, allowing them to view assigned tasks, priorities, deadlines, and required responses.
- Stakeholders update progress and report completed actions, enabling authorities to monitor implementation and coordinate responses in real time.



IMPLEMENTATION

FWise can be translated into practical, scalable, and collaborative real-world deployment.

Pilot Testing :Kedah , the state of rice in Malaysia .



RM5.0M agricultural flood losses in Kedah in 2025

947,621 tonnes paddy production

228,963 ha planted area

Kedah is selected as the first FWise pilot area because it is Malaysia's **largest paddy-producing region** while facing **significant exposure to flood and other climate-related risks**. Its high concentration of rice production, agricultural stakeholders and documented flood vulnerability provides a strong real-world environment to test FWise's data-to-decision workflow.

Source :DOSM

From April 2026 – August 2026 …

Step 1 : Build data and industries network



Before Project kick-off , FWise collaborates with **data-providing agencies** that already monitor the climate, water, agriculture, disasters, and food system.
After this , FWise brings in primary producers , food industries and supply-chain companies that depend on this information
Our Partner : MET Malaysia,JPS Kedah , MADA , Department of Agricultural,NADMA,KPKM,KPDN,DOSM

Step 2 : Deploy FWise Stations in Kedah

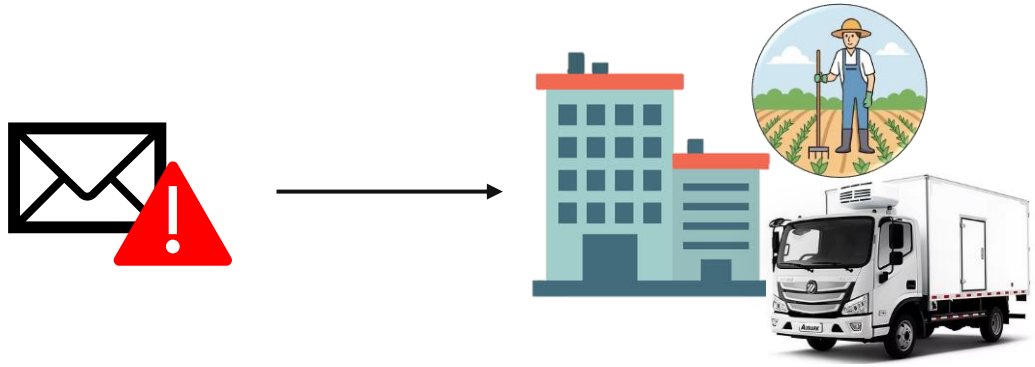


April 2026 : FWise **deploy additional AWS station/FWise Field Station** across few selected crop areas in Kedah to **fill the local information gaps** between existing regional datasets

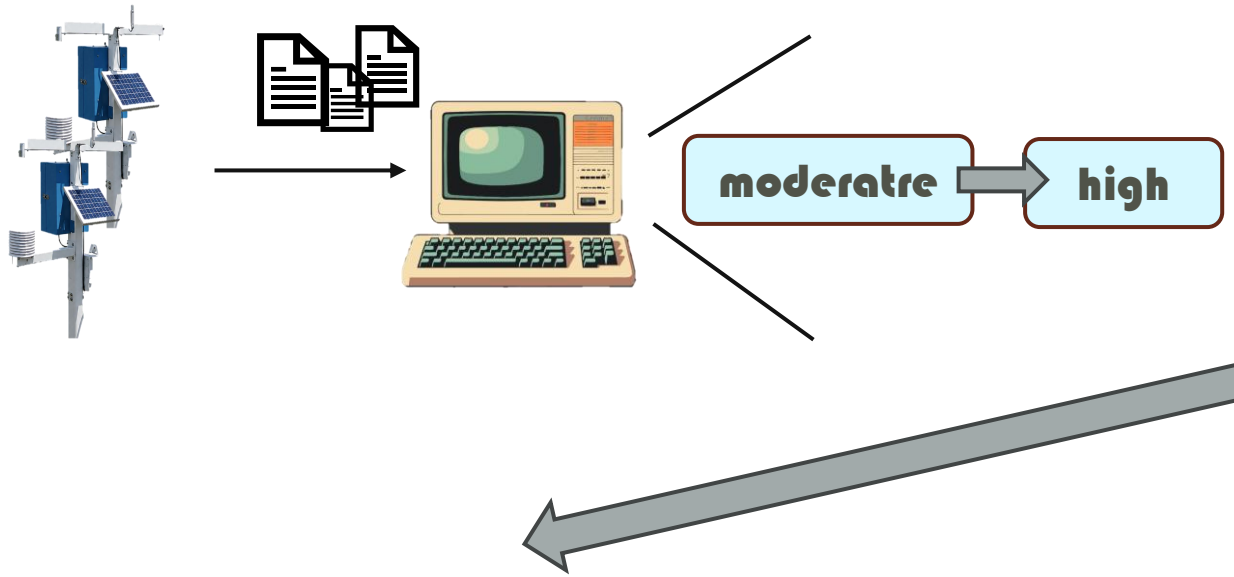
From Prediction To Action : A Successful Implementation



May 2026 : **CRIS identifies a moderate risk of severe storms and flooding occurring from mid-July to August after detecting climate conditions deviating from normal patterns.**

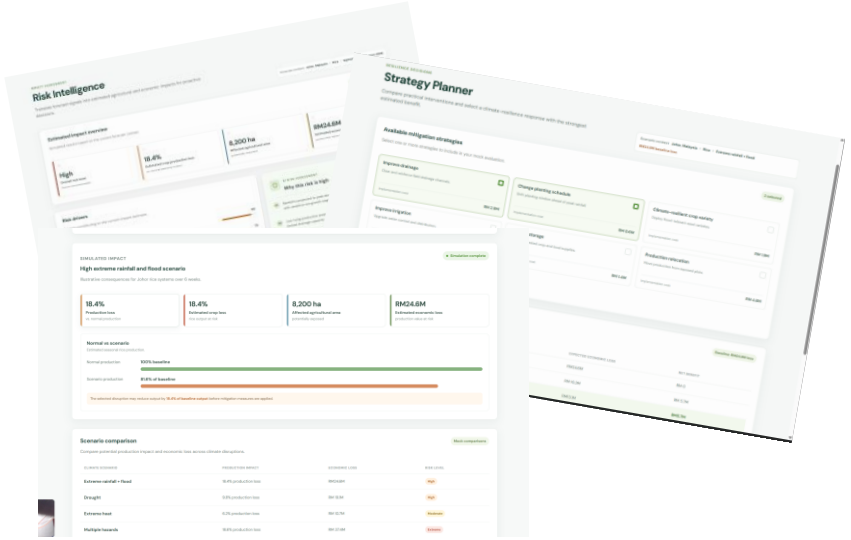


After estimating the potential production losses and economic impacts, CRIS connects the predicted disruption with registered organisations and relevant authorities, issuing early risk alerts to support timely preparedness.



Jun 2026 : **CRIS continuously monitors the latest observations from FWise field stations and incorporates updated information from participating data providers to refine the forecast.**
As multiple data sources increasingly confirm the predicted conditions, CRIS upgrades the risk assessment and formally issues a high-risk forecast for the mid-July–August period.

CRIS conducts a Food Security Stress Test to estimate potential impacts on agricultural production, food supply, and economic activity, while the Strategy Planner compares alternative response measures based on their expected costs, benefits, and loss reduction.



At the end of June ....

CRIS distributes stakeholder-specific reports, allowing relevant authorities and participating industries to select appropriate response strategies. With several weeks of lead time before the predicted event, stakeholders can begin implementing their preparedness measures.

MET Malaysia → Verify forecast → issue updated weather warning → provide 7–14 day outlook to CRIS

Persatuan Air Malaysia · Malaysian Water Association → Lower/adjust reservoir or drainage levels where operationally appropriate → inspect drainage channels → clear critical bottlenecks → prepare pumps

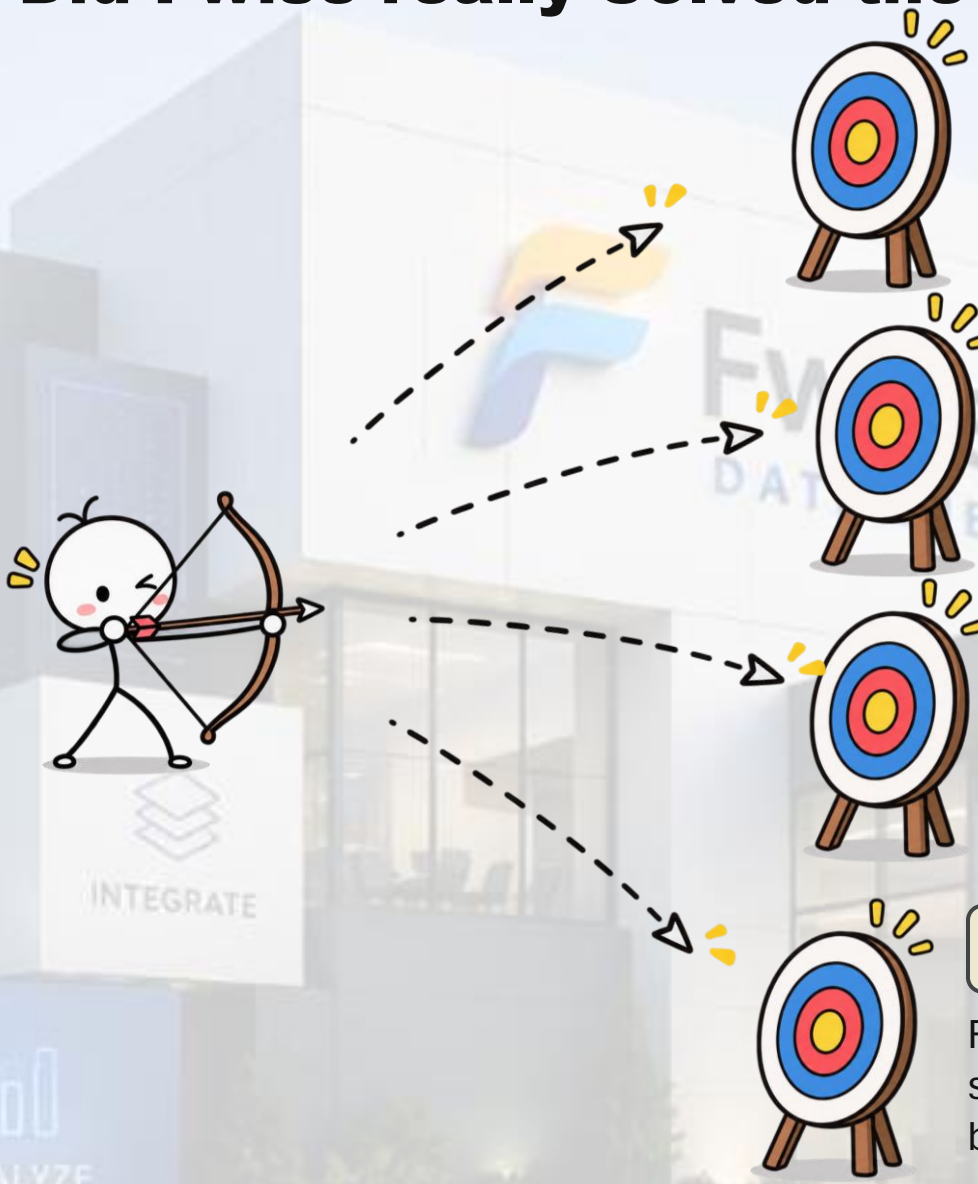
Agricultural companies → Protect machinery and inputs → adjust field operations → accelerate harvesting where feasible → activate business-continuity plans

POS Logistics
Logistics companies → Identify vulnerable routes → reserve alternative routes/vehicles → pre-position trucks → reroute planned deliveries if required

So what FWise successfully did :

- **21.7% → 7.8%** projected production loss
- **4,500 t → 829 t** residual food-supply deficit
- **RM21.8M → RM1.41M** residual economic exposure
- **Provided 4–6 weeks** preparation lead time to all relevant authorities

Did Fwise really Solved the current Problem ?



Information Fragmented

FWise establishes CRIS ,a platform where participating authorities can contribute verified hazard, agricultural, supply, demand and vulnerability data through standardised data channels.

Information Gaps

FWise field stations that continuously collect ground-level agricultural and environmental data at strategically-located area

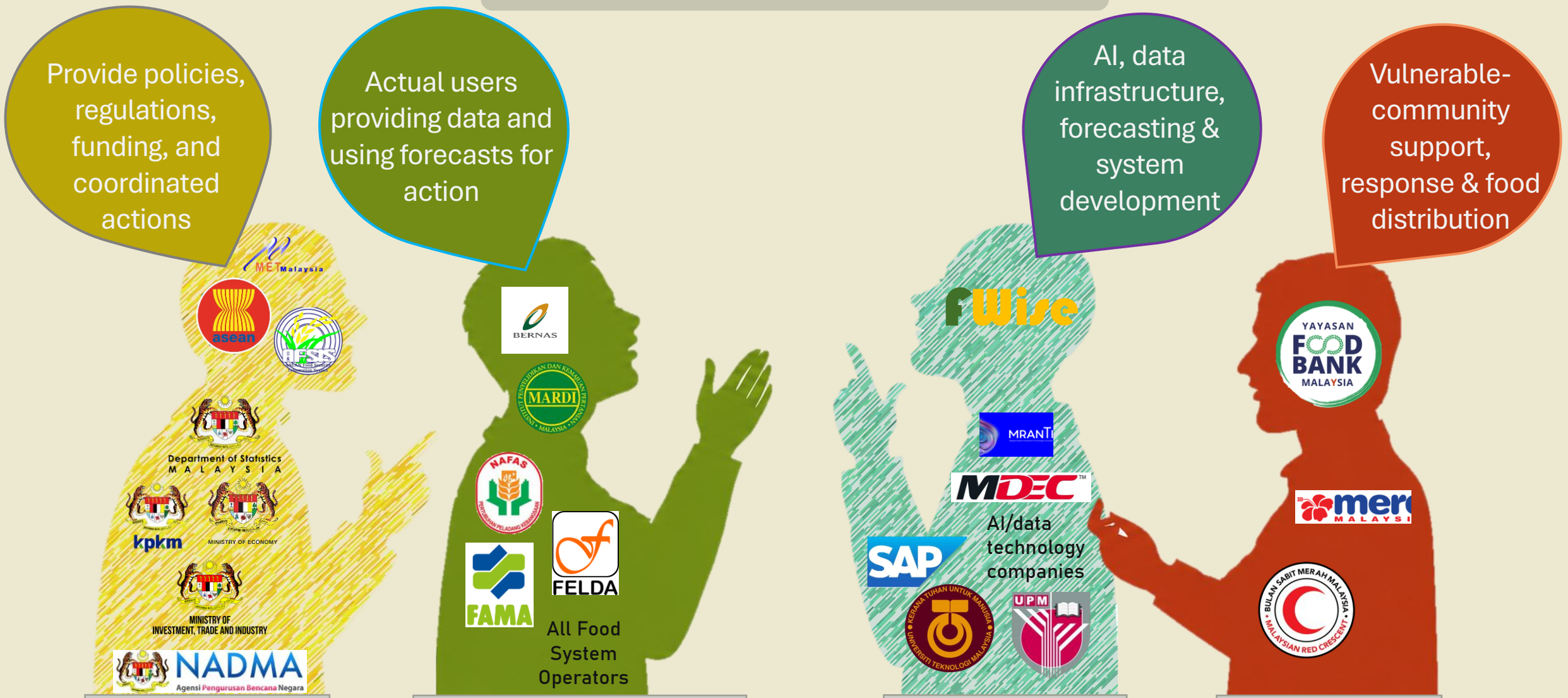
Insufficient Resilience

FWise allows resources and response measures to be mobilised before the projected shortage becomes critical.

Prediction-action Gaps

FWise's CRIS intelligence layer translates risk information into context-specific response strategies, then converts it into executable actions that can be distributed to the relevant authorities.

Skateholders Engagement



SDGs Alignment



Zero Hunger

2.1 Access to safe, nutritious and sufficient food
2.3 Double agricultural productivity and incomes of small-scale food producers
2.4 Sustainable food production and resilient agricultural practices



Climate Action

13.1 Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries



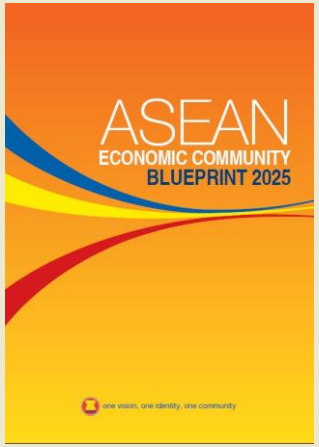
Responsible Consumption and Production

12.3 reduce food losses along production and supply chains



Sustainable Cities and Communities

11.5 decrease the direct economic losses relative to global gross domestic product caused by disasters



5.3.3 Esnure integrated food security information

Then , what are the next steps ?



Connecting Fwise with Asean Country and establish :

- Cross-border data-sharing partnerships
- Common data standards
- Climate and disaster information exchange
- Regional food-production monitoring
- Cross-border supply-chain visibility



Phase 1

- Deploy FWise facilities and field stations across Malaysia.
- Expand partnerships with all relevant and reliable agencies.
- Increase participation from farmers, SMEs and major agricultural or food companies.
- Establish national data-sharing, authentication and governance standards.
- Validate CRIS forecasting and preparedness performance across different regions.

Phase 2



Phase 3

- Expand physical facilities across ASEAN regions.
- Connect national climate, agriculture,disaster information.
- Monitor cross-border food-production risks.
- Identify regional supply shortage
- Simulate disruption scenarios involving multiple countries.
- Coordinate alternative food-supply and logistics strategies.

Final Stage

Building ASEAN Resilience mechanisms :

FWise support regional early warning .regional food supply contingency , emergency supply coordination , cross border response strategics and regional stress testing

OUR *Final Goals* :

" A CLIMATE-RESILIENT ASEAN WITH SUFFICIENT, SAFE AND ACCESSIBLE FOOD FOR ALL BY 2040."

FWise



Thank You

References

1. FAO et al. (2026), The State of Food Security and Nutrition in the World 2026 https://www.fao.org/publications/fao-flagship-publications/the-state-of-food-security-and-nutrition-in-the-world/en
2. Food and Agriculture Organization of the United Nations. (2026). *Indicator 2.1.2: Prevalence of moderate or severe food insecurity in the population, based on the Food Insecurity Experience Scale*. https://www.fao.org/sustainable-development-goals-data-portal/data/indicators/212-prevalence-of-moderate-or-severe-food-insecurity-in-the-population-based-on-the-food-insecurity-experience-scale/en
3. Southeast Asian Regional Center for Graduate Study and Research in Agriculture. (2026, July 29). *ASEAN Food, Agriculture and Forestry Sectoral Plan 2026–2030: Policy insights for a more resilient and sustainable Southeast Asia*. SEARCA Knowledge Platform https://knowledgeplatform.searca.org/insight/asean-food-agriculture-and-forestry-sectoral-plan-2026-2030-policy-insights-for-a-more-resilient-and-sustainable-southeast-asia/
4. Food and Agriculture Organization of the United Nations, International Fund for Agricultural Development, United Nations Children's Fund, World Food Programme, & World Health Organization. (2025). *The state of food security and nutrition in the world 2025: Addressing high food price inflation for food security and nutrition*. FAO. https://doi.org/10.4060/cd6008en
5. Department of Statistics Malaysia. (2026). *Performance of Malaysia's agrofood sector from a production perspective*.https://www.dosm.gov.my/uploads/release-content/file_20260327095331.pdf
6. Department of Statistics Malaysia. (2025). *Supply and utilization accounts selected agricultural commodities, Malaysia, 2020–2024* https://www.dosm.gov.my/portal-main/release-content/supply-and-utilization-accounts-selected-agricultural-commodities-malaysia-2020-2024
7. Malaysian Agricultural Research and Development Institute. (n.d.). *Engineering: e-SIAAP—early warning system application of warfare pest attacks*. https://www.mardi.gov.my/en/research-focus/engineering.html

8. Malaysian Agricultural Research and Development Institute. (n.d.). *Rice research*. https://www.mardi.gov.my/en/research-focus/rice.html

9. Department of Statistics Malaysia. (2025). *Supply and utilization accounts selected agricultural commodities, Malaysia, 2020–2024*. https://www.dosm.gov.my/uploads/release-content/file_20260311090217.pdf

10 . Department of Statistics Malaysia. (2026). *Special release: Household food waste in Malaysia 2025*. https://www.dosm.gov.my/portal-main/release-content/special-release-household-food-waste-in-malaysia-2025

11. Department of Statistics Malaysia. (2025). *Special report on impact of floods in Malaysia, 2024*. https://www.dosm.gov.my/portal-main/release-content/special-report-on-impact-of-floods-in-malaysia-2024

References

12. Department of Statistics Malaysia. (2026). *Special report on impact of floods in Malaysia, 2025*. https://www.dosm.gov.my/portal-main/release-content/special-report-on-impact-of-floods-in-malaysia2025
13. World Bank. (2024). *Malaysia economic monitor, October 2024: Farming the future—Harvesting Malaysia's agricultural resilience through digital technologies*. World Bank. https://openknowledge.worldbank.org/server/api/core/bitstreams/d1fac5c5-b948-417e-95cf-b6036f3a6756/content
14. World Bank. (n.d.). *Malaysia—Climatology (ERA5)*. Climate Change Knowledge Portal. https://climateknowledgeportal.worldbank.org/country/malaysia/era5-historical
15. Department of Statistics Malaysia. (2025). *Special report on impact of floods in Malaysia, 2024*. https://www.dosm.gov.my/portal-main/release-content/special-report-on-impact-of-floods-in-malaysia-2024
16. World Bank. (2023). *Malaysia climate data catalogue assessment*. https://documents1.worldbank.org/curated/en/099511006272312389/pdf/IDU083fda98f0896b04dc50bcc105bfd4707dbe6.pdf
17. OECD. (2024). *OECD economic surveys: Malaysia 2024*. OECD Publishing. https://doi.org/10.1787/e45ca31a-en
18. World Bank. (2024). *Malaysia economic monitor: Farming the future—Harvesting Malaysia's agricultural resilience through digital technologies*. World Bank https://www.worldbank.org/en/country/malaysia/publication/farming-the-future-harvesting-malaysia-s-agricultural-resilience-through-digital-technologies
19. **Department of Statistics Malaysia. (2025).** *Special report on impact of floods in Malaysia, 2024.* https://www.dosm.gov.my/portal-main/release-content/special-report-on-impact-of-floods-in-malaysia-2024
20. Department of Statistics Malaysia. (2025). *Selected agricultural indicators, Malaysia, 2024*. https://www.dosm.gov.my/portal-main/release-content/selected-agricultural-indicators-malaysia-2024
21. Department of Statistics Malaysia. (2026). *Supply and utilization accounts: Selected agricultural commodities, 2020–2024*. https://www.dosm.gov.my/uploads/release-content/file_20260311085907.pdf

References

22. Department of Statistics Malaysia. (2026). *Special release: Household food waste in Malaysia 2025*. https://www.dosm.gov.my/site/downloadrelease?id=special-release-household-food-waste-in-malaysia-2025&lang=English

23. Organisation for Economic Co-operation and Development. (2025). *Disaster early warning systems and private sector participation in ASEAN: Focus on the Philippines*. OECD Publishing. https://doi.org/10.1787/bc282563-en

24. World Bank. (2024). *Malaysia economic monitor: October 2024—Farming the future: Harvesting Malaysia's agricultural resilience through digital technologies*. https://openknowledge.worldbank.org/server/api/core/bitstreams/d1fac5c5-b948-417e-95cf-b6036f3a6756/content

25. ASEAN Secretariat. (2020). *Ready for the dry years: Building resilience to drought in South-East Asia with a focus on Cambodia, Lao People's Democratic Republic, Myanmar and Viet Nam: 2020 update*. https://asean.org/book/ready-for-the-dry-years-building-resilience-to-drought-in-south-east-asia-with-a-focus-on-cambodia-lao-peoples-democratic-republic-myanmar-and-viet-nam-2020-update/

26. Food and Agriculture Organization of the United Nations, International Fund for Agricultural Development, United Nations Children's Fund, World Food Programme, & World Health Organization. (2025). *The state of food security and nutrition in the world 2025: Addressing high food price inflation for food security and nutrition*. https://www.fao.org/agrifood-economics/publications/detail/en/c/1740904/

27. World Bank. (2024). *Flood risk management: Leveraging finance for business resilience in Malaysia*. https://www.worldbank.org/en/country/malaysia/publication/flood-risk-management-leveraging-finance-for-business-resilience-in-malaysia

28. National Disaster Management Agency. (2025, July 15). *Floods wiped out RM158m of padi in 2024*. https://www.nadma.gov.my/bm/media-2/berita/4810-floods-wiped-out-rm158m-of-padi-in-2024

29. Department of Statistics Malaysia. (2026). *Special report on impact of floods in Malaysia, 2025*. https://www.dosm.gov.my/portal-main/release-content/special-report-on-impact-of-floods-in-malaysia2025

30. Malaysian Meteorological Department. (n.d.). *Meteorological instruments*. https://www.met.gov.my/en/pendidikan/peralatan-meteorologi/

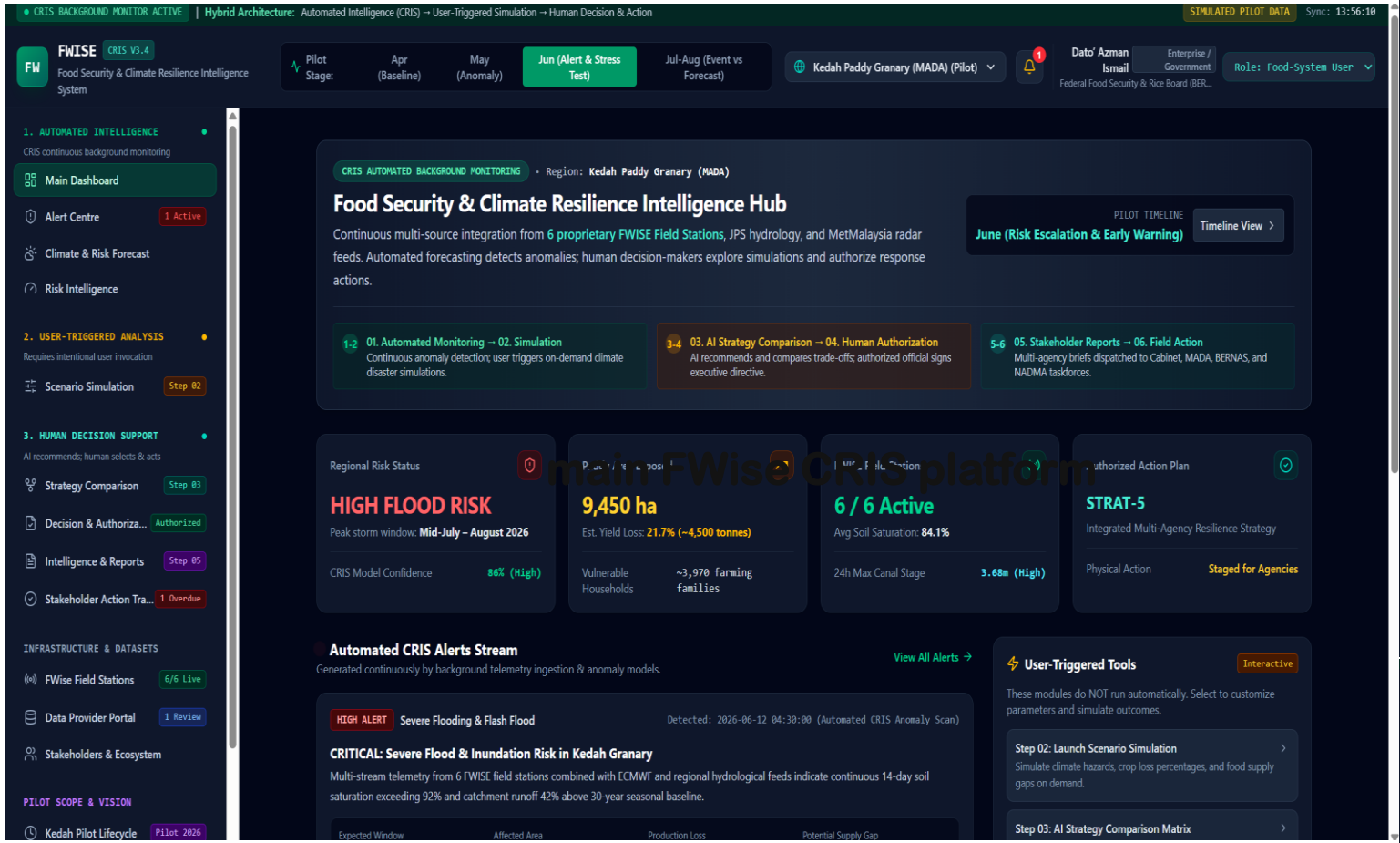
31. Department of Statistics Malaysia. (2026). *Agriculture Census 2024 report: Kedah* https://www.dosm.gov.my/portal-main/release-content/agriculture-census-2024-report-kedah

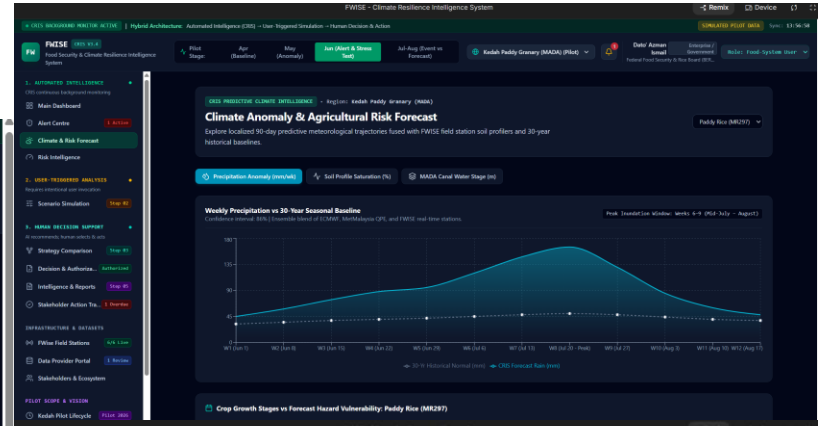
32. Department of Statistics Malaysia. (2026). *Special report on impact of floods in Malaysia, 2025*. https://www.dosm.gov.my/portal-main/release-content/special-report-on-impact-of-floods-in-malaysia2025

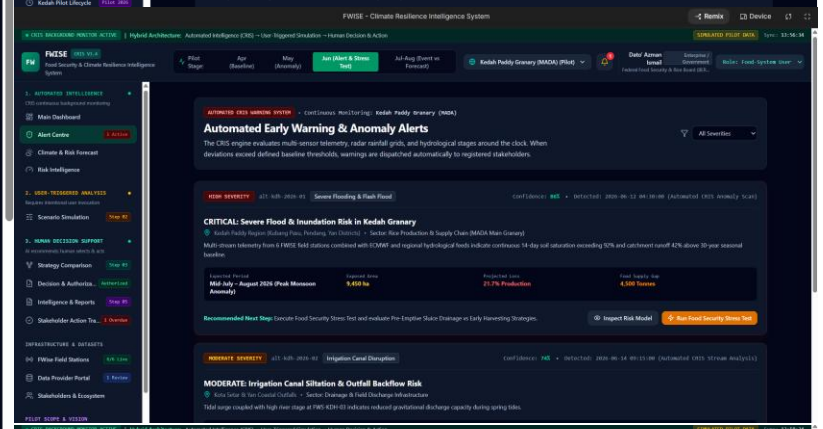
33. Ideris, M. M., Amin, M. Z. M., Cai, G. Y., & Adnan, N. H. M. (2014). *Climate change impacts on agriculture water management*. National Water Research Institute of Malaysia (NAHRIM). https://digital.nahrim.gov.my/portal-main/publication-details?id=15

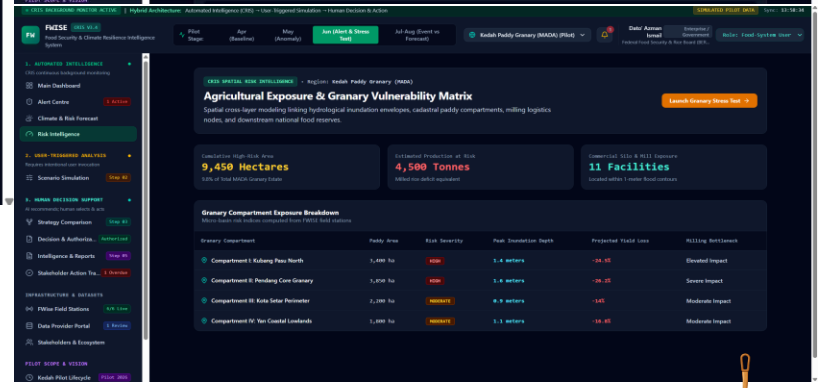
34. Department of Statistics Malaysia. (2025). *Supply and utilization accounts selected agricultural commodities, Malaysia, 2020–2024*. Department of Statistics Malaysia. https://www.dosm.gov.my/portal-main/release-content/supply-and-utilization-accounts-selected-agricultural-commodities-malaysia-2020-2024?

Appendix A :main FWise CRIS system





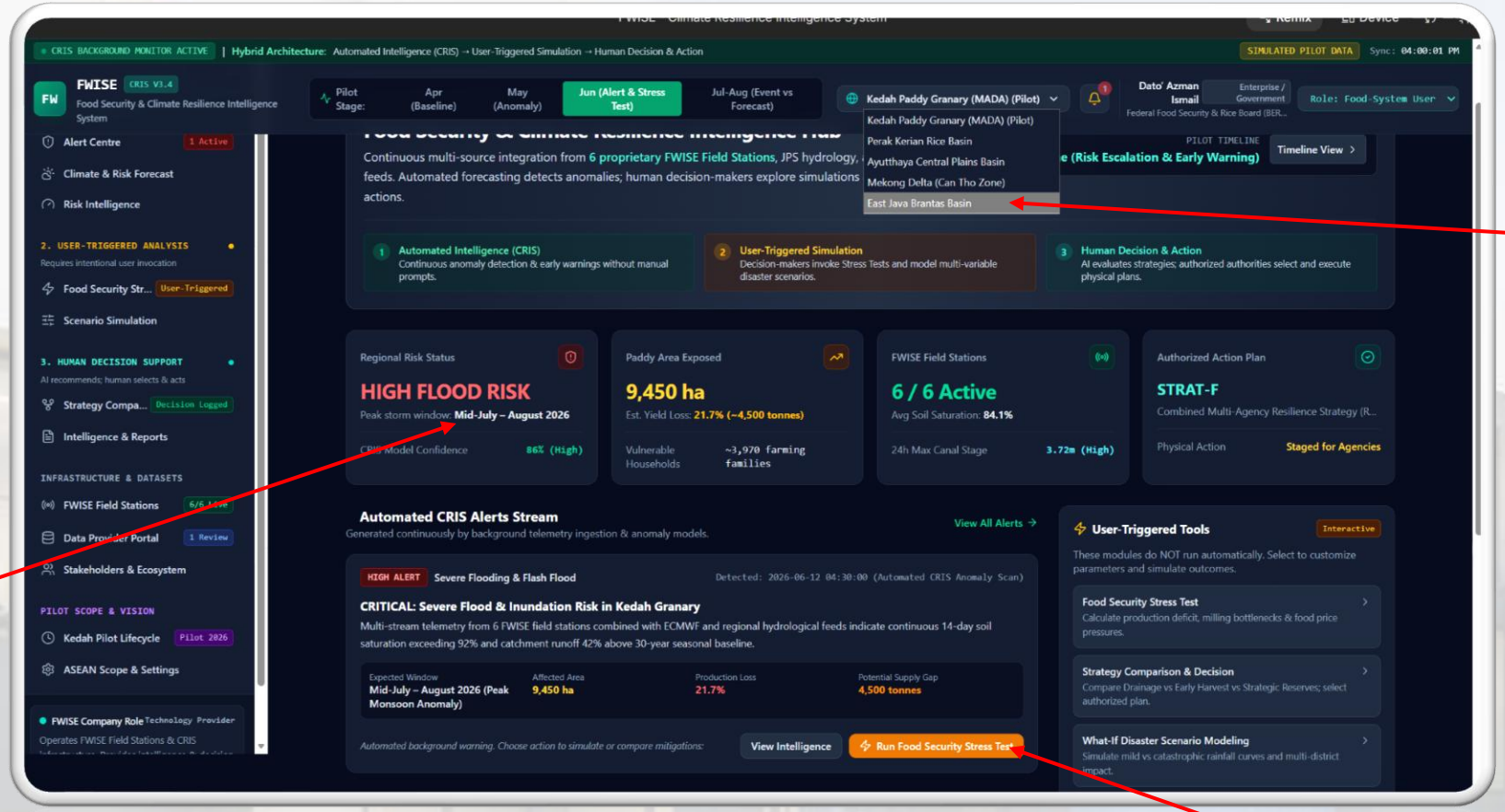




main FWise CRIS platform

For Data Providers : **main FWise CRIS platform**.

SYSTEM DASHBOARD



1.Choose piloting area

2.Check latest forecast information

3. Run the stress test for more details information

CRIS : FOOD SECURITY STRESS TESTING ENGINE

CURRENT CRIS DISASTER ASSESSMENT

+ FWise Field Stations (FWS 01-06)
+ JPS & MetMalaysia Radar Telemetry

Severe Monsoon Deluge & Tidal Lock

Automatically generated from the latest CRIS alert and pilot-area telemetry. This real-world baseline serves as the official unmitigated hazard model for national response strategy evaluations.

Affected Area	Projected Loss	Supply Deficit	Event Duration
9,450 ha	21.7%	4,500 t	12 days
9.8% of MADA main granary	4,500 tonnes standing paddy	Consumer & milling shortfall	Prolonged tidal lock ponding

Rainfall Anomaly	Severity Level	Economic Exposure	Event Window
+38%	SEVERE	RM21.8M	Mid-July – Aug 2026
Above 30-year seasonal norm	Tidal Surge + River Crest	$4.85M USD equivalent	Critical grain ripening phase

HYDROLOGICAL TELEMETRY SYNTHESIS & GRANARY VULNERABILITY FOOTPRINT

Location: Kedah Paddy Granary (MADA Blocks I–IV)

HYDRAULIC CAUSE:
Synchronous high river discharge and coastal astronomical high tides blocking natural estuary outfalls across Sungai Kedah & Sungai Muda rivermouths. Water level exceeds +2.4m bund elevation.

FARMING SMALLHOLDERS:
3,970 Farm Households
Registered MADA farmers facing total loss of ripening standing crops without pre-emptive harvest.

INDUSTRIAL RICE MILLS:
11 Co
Drying opera
roadway acc

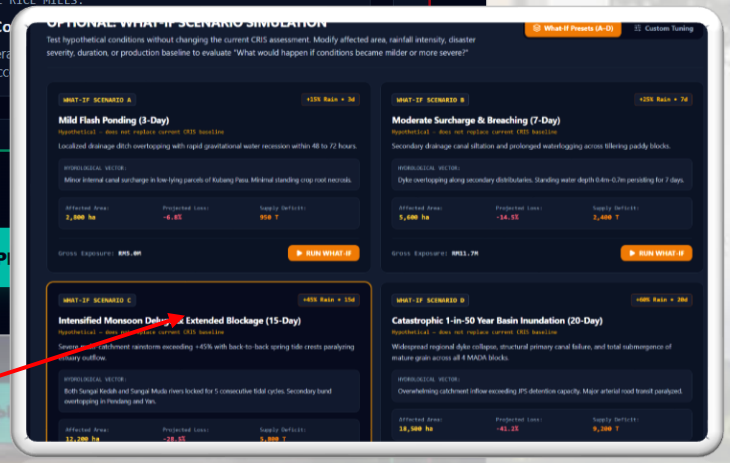
ACTIVE BASELINE READY FOR DECISION SUPPORT

Severe Monsoon Deluge & Tidal Lock is the Active Crisis Baseline.

Execute automated decision support against this active baseline to evaluate response strategies, generate CRIS recommendations, and prepare executive directives.

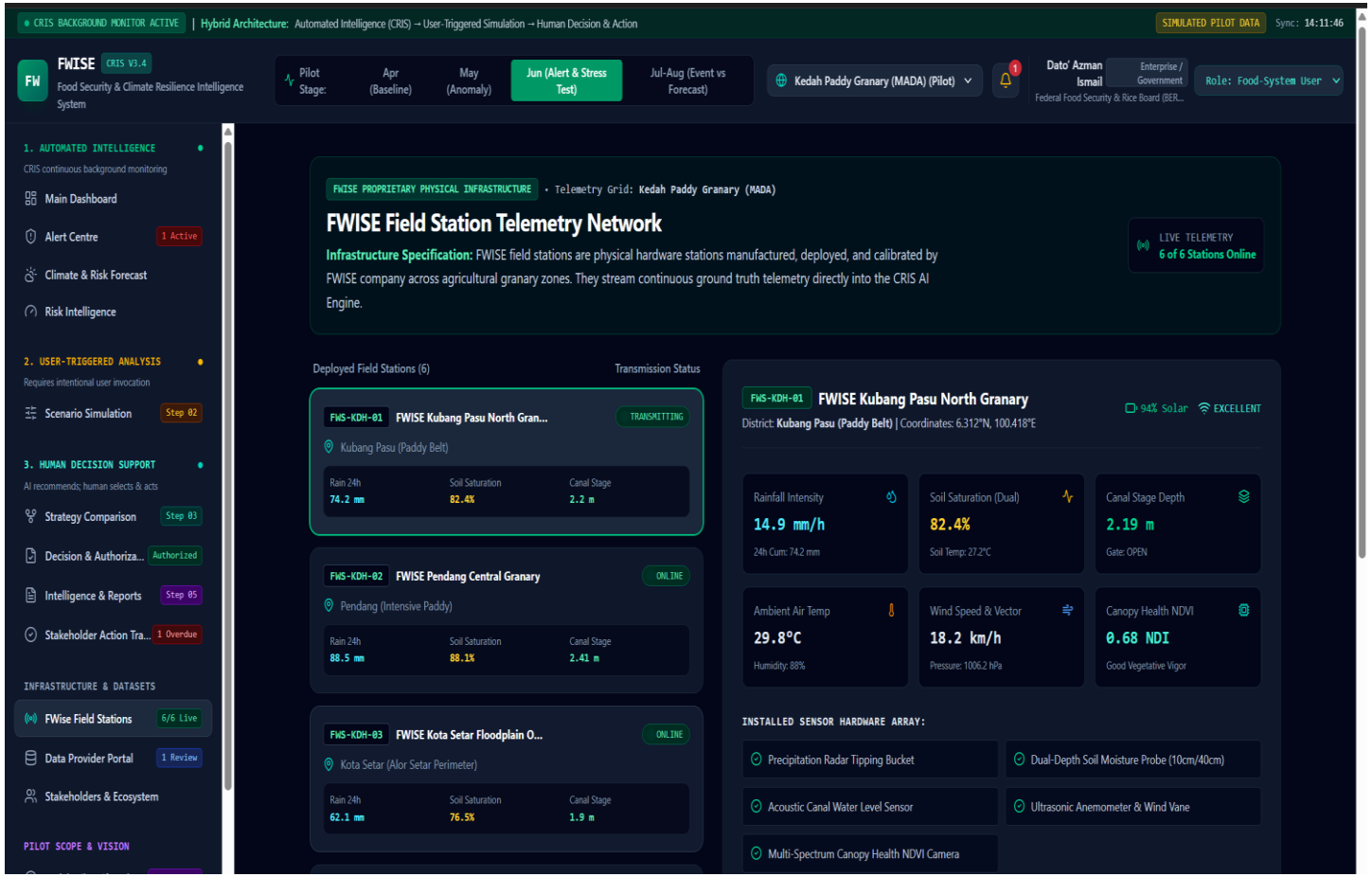
GENERATE DECISION SUP

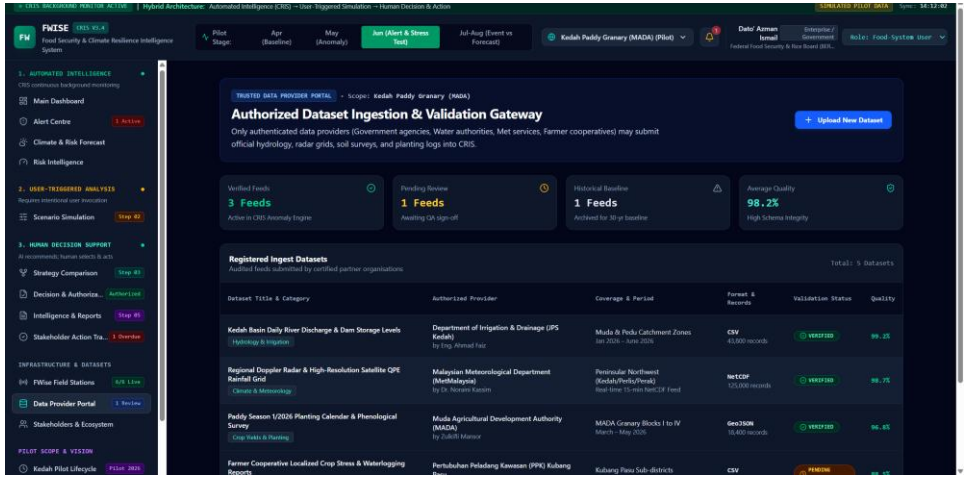
Results Displayment

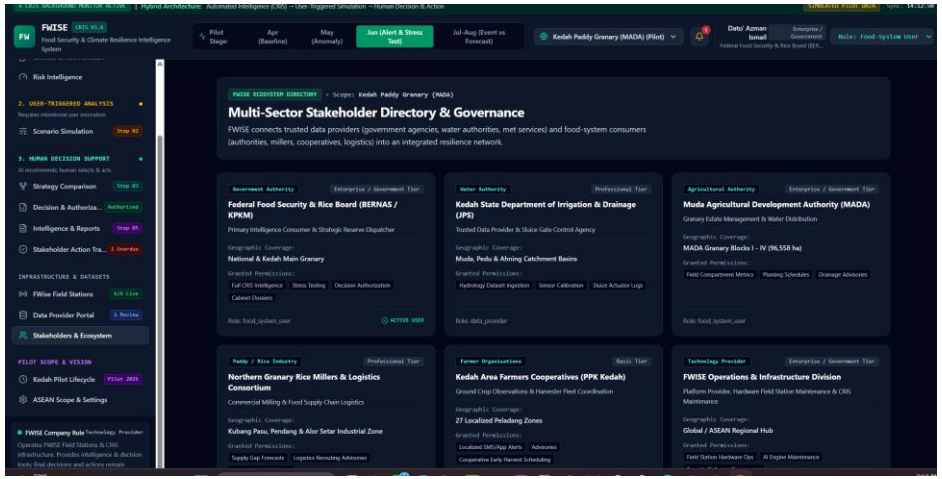


FWise also support different scenario simulation

Appendix A2 – Fwise system : Datasets & Infrastructure

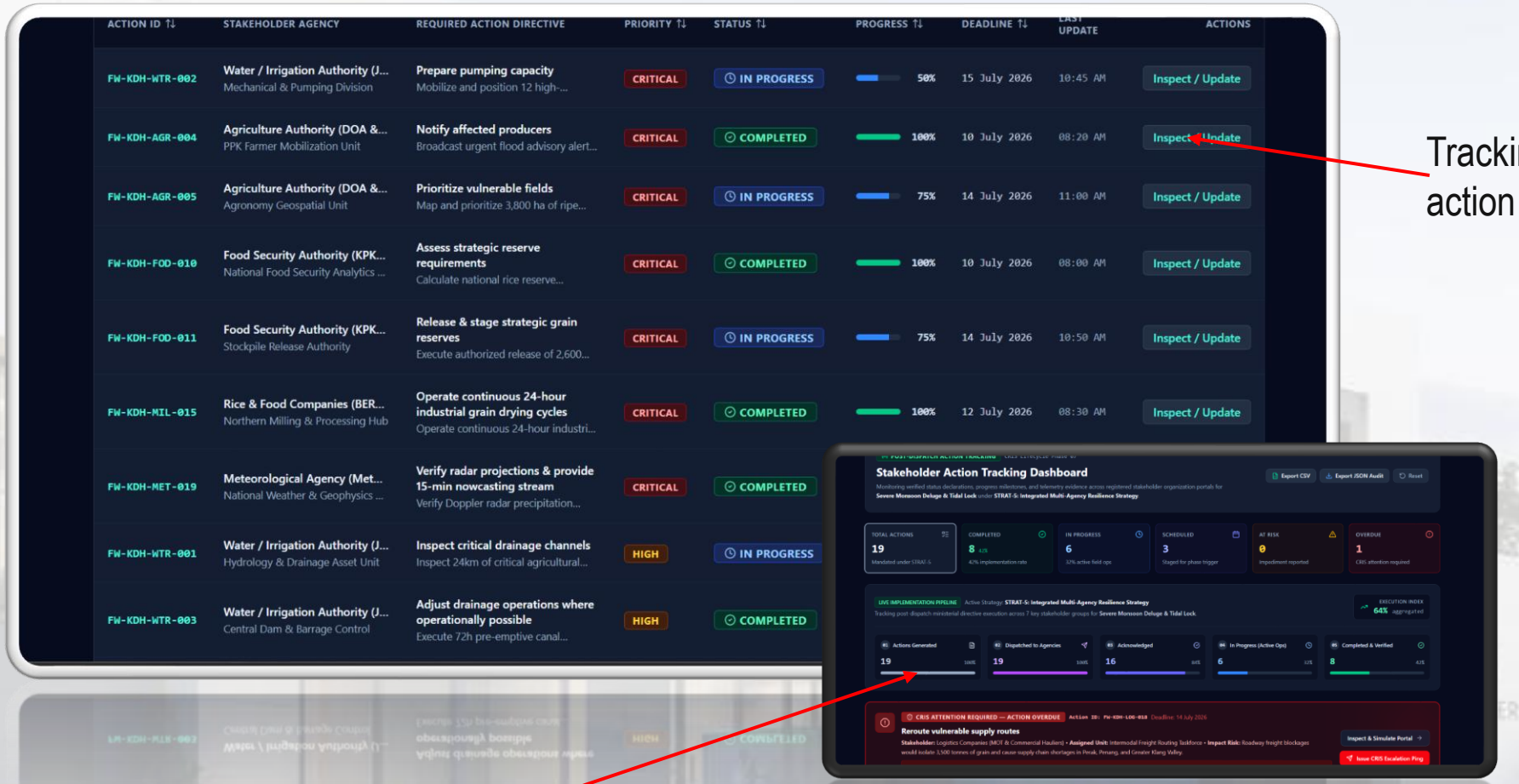






"How do we verify that the recommended actions have been completed?"

STAKEHOLDER ACTION TRACKING

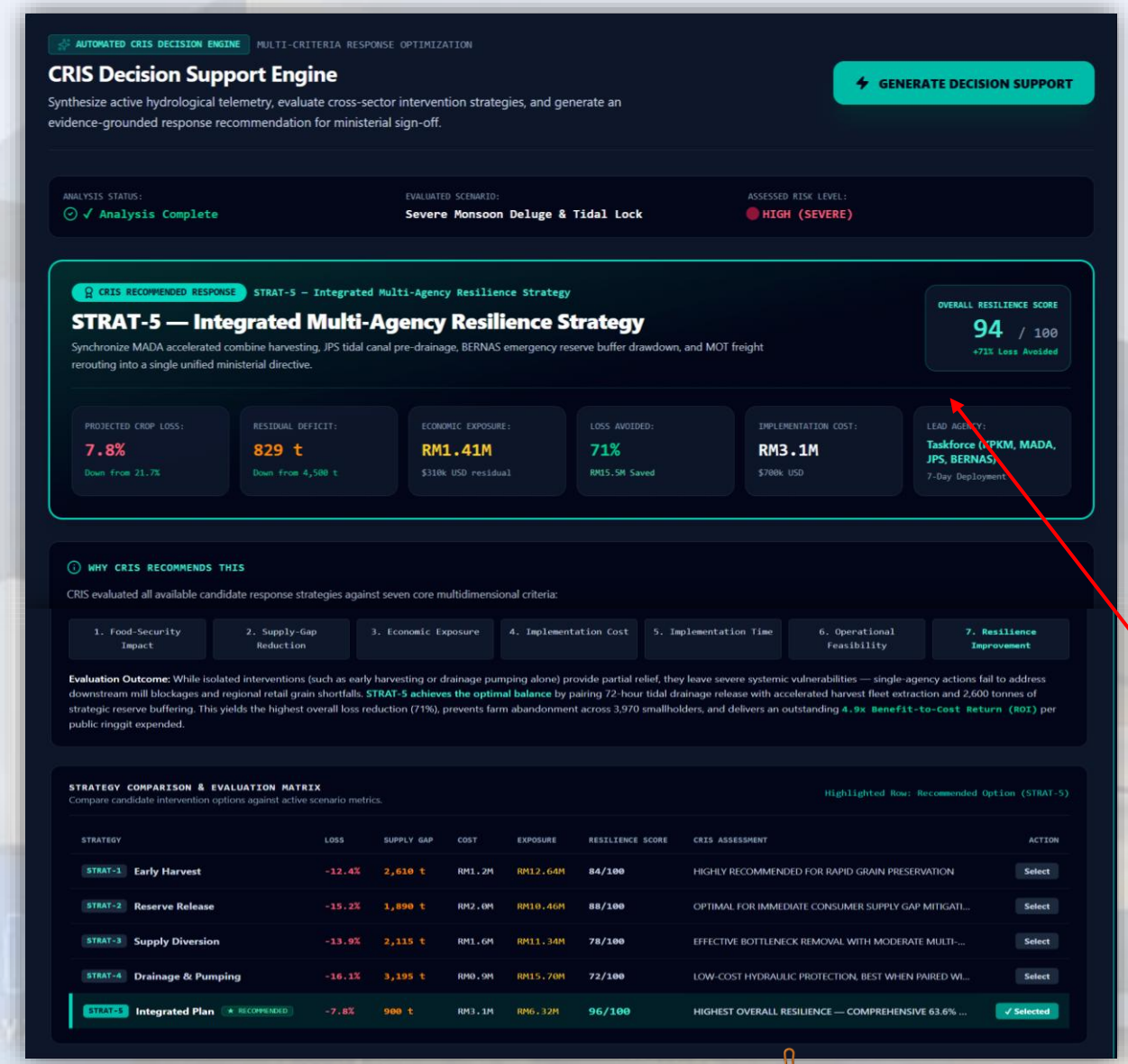


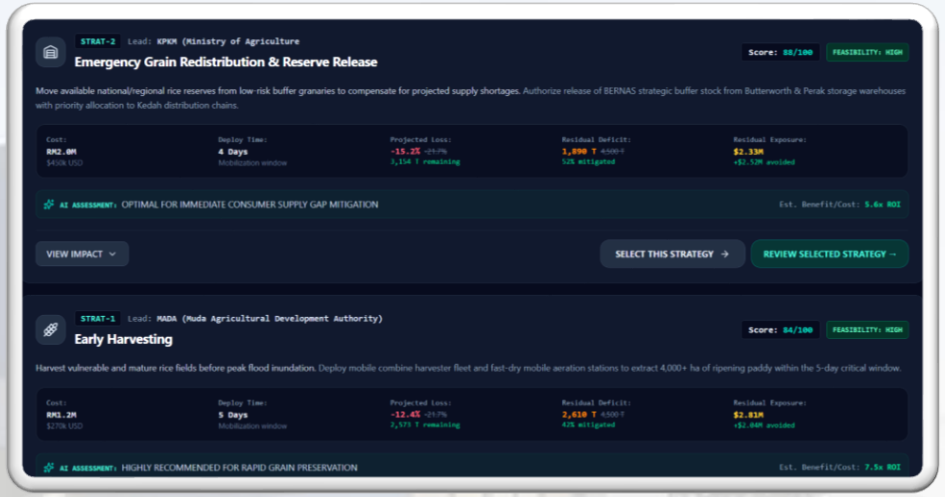
ACTION ID	STAKEHOLDER AGENCY	REQUIRED ACTION DIRECTIVE	PRIORITY	STATUS	PROGRESS	DEADLINE	LAST UPDATE	ACTIONS
FW-KDH-WTR-002	Water / Irrigation Authority (J... Mechanical & Pumping Division	Prepare pumping capacity Mobilize and position 12 high-...	CRITICAL	IN PROGRESS	50%	15 July 2026	10:45 AM	Inspect / Update
FW-KDH-AGR-004	Agriculture Authority (DOA &... PPK Farmer Mobilization Unit	Notify affected producers Broadcast urgent flood advisory alert...	CRITICAL	COMPLETED	100%	10 July 2026	08:20 AM	Inspect / Update
FW-KDH-AGR-005	Agriculture Authority (DOA &... Agronomy Geospatial Unit	Prioritize vulnerable fields Map and prioritize 3,800 ha of ripe...	CRITICAL	IN PROGRESS	75%	14 July 2026	11:00 AM	Inspect / Update
FW-KDH-FOD-010	Food Security Authority (KPK... National Food Security Analytics ...	Assess strategic reserve requirements Calculate national rice reserve...	CRITICAL	COMPLETED	100%	10 July 2026	08:00 AM	Inspect / Update
FW-KDH-FOD-011	Food Security Authority (KPK... Stockpile Release Authority	Release & stage strategic grain reserves Execute authorized release of 2,600...	CRITICAL	IN PROGRESS	75%	14 July 2026	10:50 AM	Inspect / Update
FW-KDH-MIL-015	Rice & Food Companies (BER... Northern Milling & Processing Hub	Operate continuous 24-hour industrial grain drying cycles Operate continuous 24-hour industri...	CRITICAL	COMPLETED	100%	12 July 2026	08:30 AM	Inspect / Update
FW-KDH-MET-019	Meteorological Agency (Met... National Weather & Geophysics ...	Verify radar projections & provide 15-min nowcasting stream Verify Doppler radar precipitation...	CRITICAL	COMPLETED				
FW-KDH-WTR-001	Water / Irrigation Authority (J... Hydrology & Drainage Asset Unit	Inspect critical drainage channels Inspect 24km of critical agricultural...	HIGH	IN PROGRESS				
FW-KDH-WTR-003	Water / Irrigation Authority (J... Central Dam & Barrage Control	Adjust drainage operations where operationally possible Execute 72h pre-emptive canal...	HIGH	COMPLETED				

Tracking each action progress

Monitoring overall action progress

CRIS :DECISION SUPPORT ENGINE

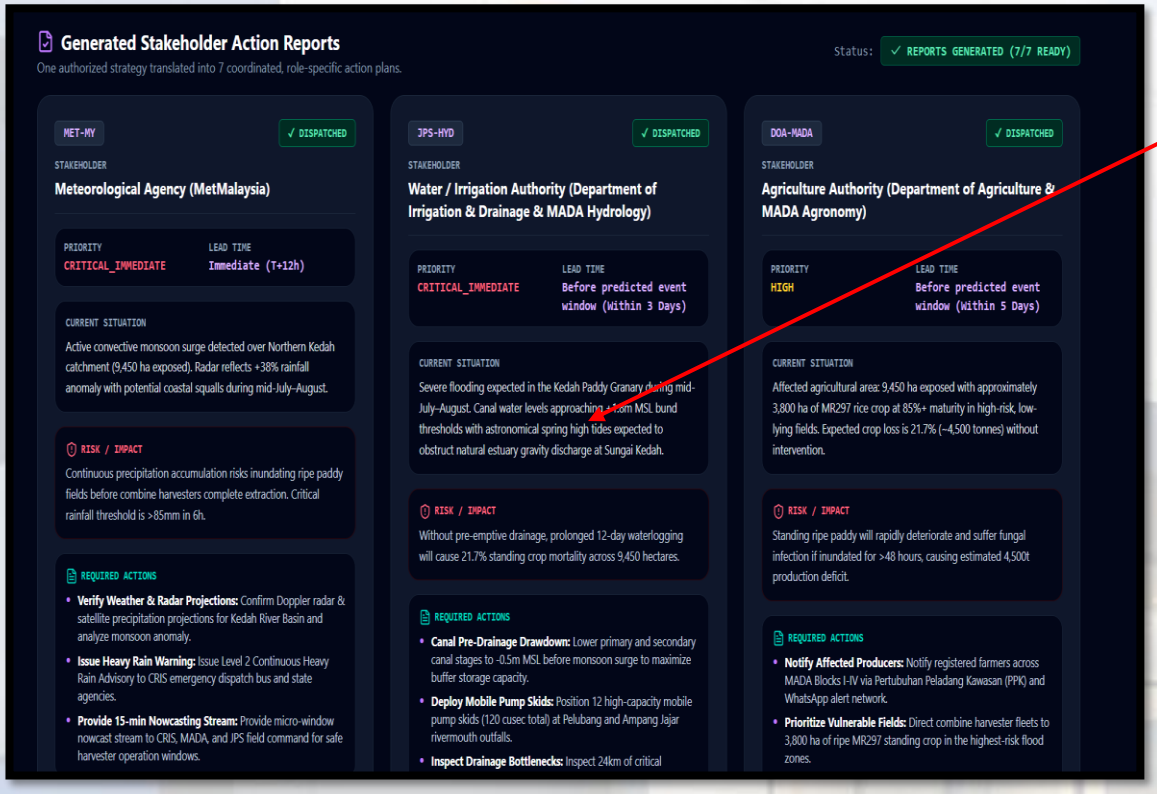




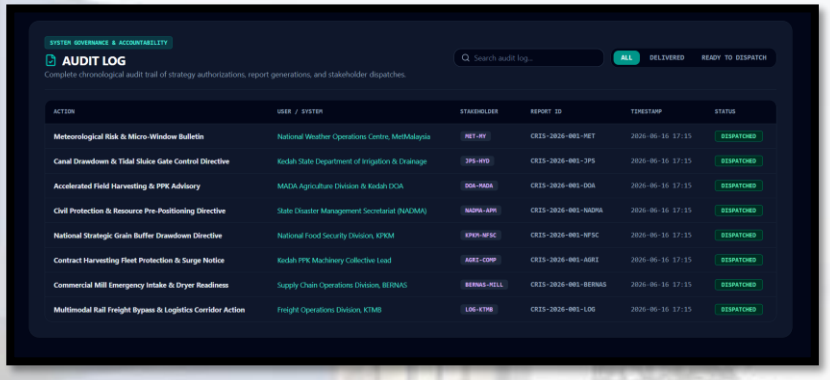
AI-suggested best option

2.Users can instantly compare all response strategies and select the most effective option.

AUTOMATED REPORT DISTRIBUTION

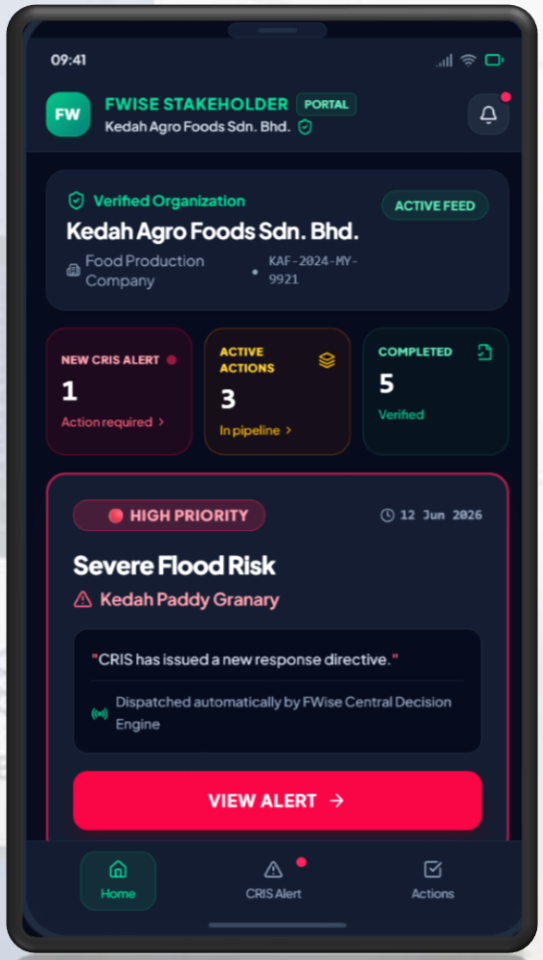


Action Details

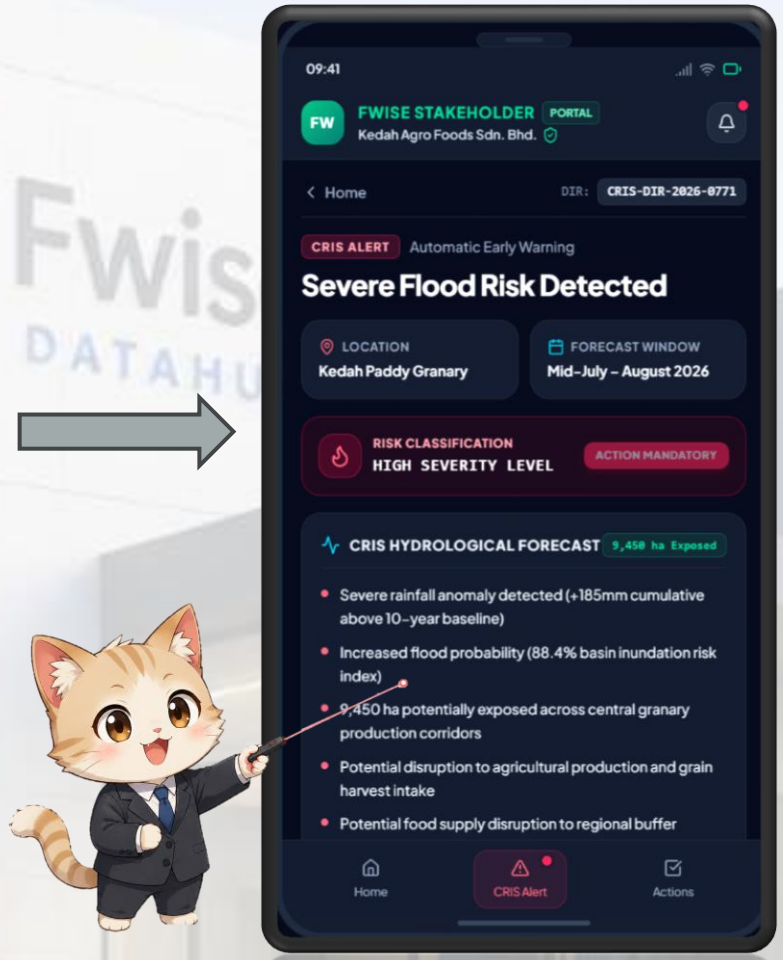


CRIS automatically translates risk predictions into tailored action reports for the right stakeholders.

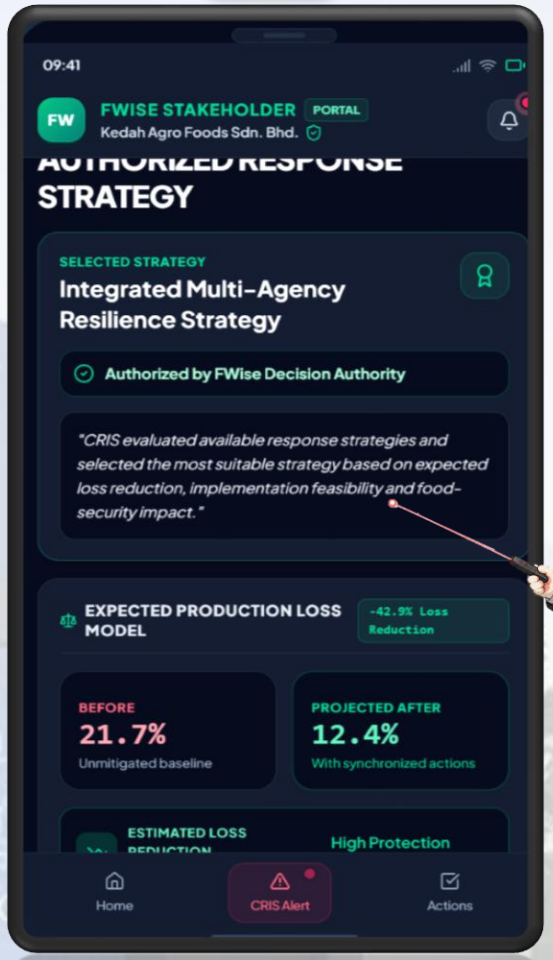
Appendix A3 : **FWise Stakeholder mobile app.**



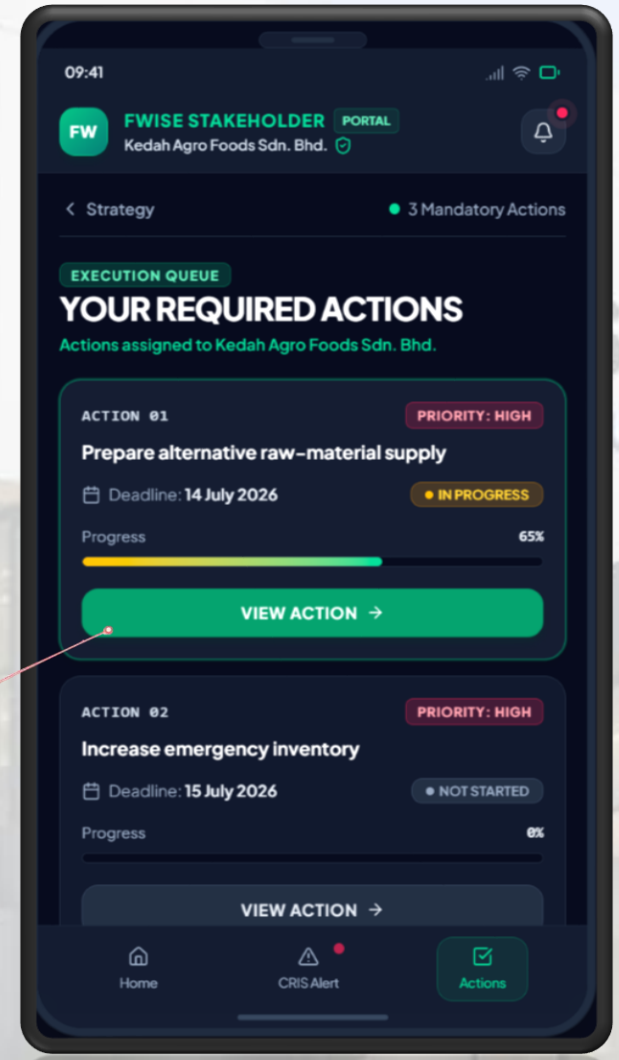
Dashboard : Receiving alerting message

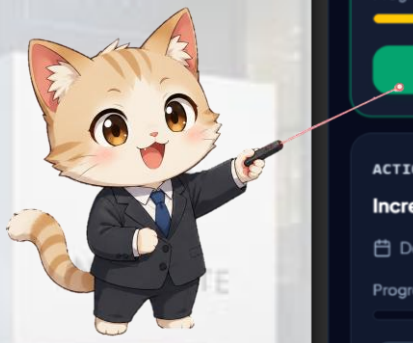


CRIS Alert: Review Forecast & Risk Details

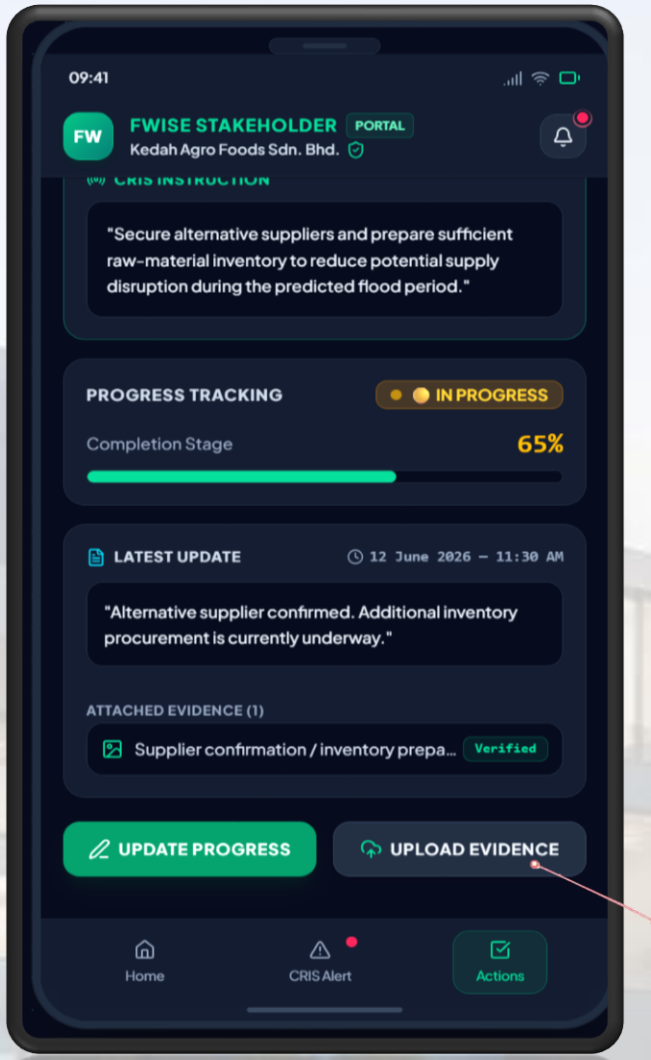


Check the authorized strategics





Tracking Required Actions





Updating Progress & Evidence

FWISE FIELD STATION – COMPONENTS

1. WIND SPEED SENSOR (ANEMOMETER)



Measures wind speed, gust, and average wind speed.

2. WIND DIRECTION SENSOR (WIND VANE)



Measures wind direction and direction changes.

3. AIR TEMPERATURE SENSOR (RADIATION SHIELD)



Measures air temperature, minimum, maximum, and average.

4. RELATIVE HUMIDITY SENSOR (IN RADIATION SHIELD)



Measures relative humidity and helps calculate dew point and VPD.

5. RAIN GAUGE / PRECIPITATION SENSOR



Measures reinfall amount, intensity, and accumulation.

6. SOLAR RADIATION SENSOR (PYRANOMETER)



Measures solar radiation (W/m^2) for crop growth and evapotranspiration.

7. BAROMETRIC PRESSURE SENSOR



Measures atmospheric pressure (hPa) for weather trend analysis.

8. SOIL MONITORING SENSOR (ARRAY)



Measures soil moisture, soil temperature, and optional EC/salinity.

9. WATER LEVEL / FLOOD SENSOR



Measures water level, rate of rise, and flood threshold.

10. DATA LOGGER + COMMUNICATION UNIT



Collects, stores, processes, and transmits all sensor data to FWISE.

11. COMMUNICATION SYSTEM (ANTENNA)



Transmits data via 4G/5G, LoRaWAN, or Satellite.

12. SOLAR POWER SYSTEM



Provides power through solar panel, charge controller, and battery.

13. STATION HEALTH INDICATOR (STATUS LIGHT / DISPLAY)



Shows station health status, battery level, and system alerts.

14. MOUNTING POLE & STRUCTURE



Supports all components and ensures proper installation.

SUMMARY OF WHAT THE STATION DOES

- Collects real-time environmental and agricultural data.
- Detects weather, crop, and flood risks.
- Transmits data to FWISE platform.
- Supports forecasting, risk assessment, and early warning.
- Helps improve agricultural decision-making and disaster preparedness.

FWise Workflow



FWise collects localized climate, environmental, agricultural, and hazard data from field stations and sending it to Fwise CRIS system

CRIS analyses the integrated data to detect risks, forecast potential threats and assess their implications for food security.



If there are the unusual data have been detected,Fwise will keep tracking developing data and updates the risk assessment.

When a high-risk condition is detected, FWise automatically issues an alert with a comprehensive strategic response plan, outlining the recommended actions, priorities and measures for relevant stakeholders to implement.







To track whether the actions have been implemented, relevant stakeholders are required to submit a comprehensive action report through the FWise mobile app. The submitted reports allow the system to monitor implementation progress, verify completed actions, and provide authorities with up-to-date response status.

Appendix B : FWise Field Station — Initial Normal Reference Values

No.	Component	Parameter	Unit	Initial normal/reference value for FWise	Source
1	Wind speed	Average wind speed	**m/s**	**≈ 2.0 m/s**	DOSM — Compendium of Environment Statistics Malaysia 2025 (PDF) DOSM — Compendium of Environment Statistics https://www.met.gov.my/en/pendidikan/iklim-malaysia/ https://www.met.gov.my/en/pendidikan/peralatan-meteorologi/ https://publicinfobanjir.water.gov.my/aras-air/data-paras-air/?lang=en https://publicinfobanjir.water.gov.my/cari-station/?lang=en
2	Wind direction	Prevailing wind direction	**°**	**-**	
3	Air temperature	Mean temperature	**°C**	**≈ 27–28°C**	
4	Relative humidity	Mean RH	**%**	**≈ 80%**	
5	Rain gauge	Annual rainfall	**mm/year**	**≈ 2,500 mm/year**	
5	Rain gauge	Rainfall intensity	**mm/h**	**≈ 0–10 mm/h normal; >10 mm/h increasingly abnormal**	
6	Solar radiation	Global solar radiation	**MJ/m²/day**	**≈ 18 MJ/m²/day**	
7	Barometric pressure	Mean sea-level pressure	**hPa**	**≈ 1,010 hPa**	
8	Soil monitoring	Soil moisture	**% VWC**	**Site-calibrated; no reliable national normal**	
8	Soil monitoring	Soil temperature	**°C**	**≈ 27–30°C** for lowland tropical agricultural soils	
8	Soil monitoring	Soil EC	**dS/m**	**≈ 0–2 dS/m** as a general non-saline agricultural reference	
9	Water level	River level	**m**	**Station-specific**	
9	Water level	Flood threshold	**m**	**Use DID Alert/Warning/Danger threshold**	

Appendix B2 – Mathematic Model : Climate risk & Scenario Simulation

Climate-Risk Scoring Model

$$R_h = \sum_{i=1}^{n} w_{h,i} S_i$$

- R_h= risk score for climate hazard

S_i= standardized abnormality/risk score of environmental variable

- $w_{h,i}$ = weight of variable ii for hazard
- h= one of the possible climate hazards

Estimate production loss

$$L_{Production\ loss} = A_{affected} * Y_{baseline} * L_{rate}$$

- $A_{affected}$ = affected area (ha)
- $Y_{baseline}$ = expected baseline yield (t/ha)
- L_{rate} = estimated production-loss rate

Estimate affected area

$$A_{affected} = \sum_{i=1}^{n} A_i I_i \quad ,0 \leq I_i \leq 1$$

- A_i= area of agricultural zone(ha)
- I_i = impact indicator for zone

Appendix B2 – Mathematic Model : Decision Support System

Projected Crop Loss , $L_{loss} = \frac{\mathrm{L}_{production\ loss}}{P_{production}} x\ 100\ \%$

- $P_{production}$ = expected production without the climate event (t)
- $\mathrm{L}_{production\ loss}$ = projected quantity of production lost (t)

Residual Deficit ,$RD = Demand\ - Supply\ available$

$$\frac{RD}{Demand}\ x\ 100\ \%$$

Implementation Cost

$$C_s = \sum_{k=1}^{m} C_{k,s}$$

where:

- C_s = total implementation cost
- $C_{k,s}$ = cost of each intervention

Economic exposure , $E_s = \sum_{j=1}^{n} Q_{j,s}\ V_j$

- E_s = economic exposure under strategy ss
- Q_j = quantity of asset/resource remaining at risk
- V_j= monetary value per unit
- j = different affected assets/resources

Loss Avoided

$$L_{AS} = L_o\ -\ L_s$$

- L_{AS} = loss avoided by strategy ss
- L_o = projected loss under the baseline/no-intervention scenario
- L_s = projected loss after implementing strategy

Overall Resilience Score

$$N_{j,s}=\text{Normalize}(X_{j,s})$$

$$RS_s = \sum_{j=1}^{n} W_j\ N_{j,s}$$

- RS_s = resilience score for strategy
- $N_{j,s}$ = normalized performance of criterion
- W_j = weight assigned to criterion

Appendix C – Budget & Potential Risk Estimation : FWise Field Station

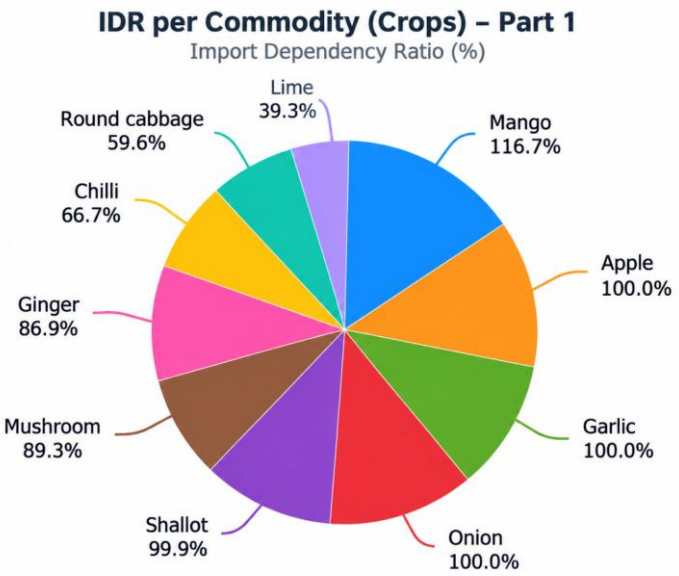
Component	Estimated cost (RM)
Air temperature & humidity sensor	1,500
Rainfall / tipping-bucket rain gauge	2,000
Wind speed & wind direction sensor	2,500
Solar radiation sensor	2,000
Atmospheric pressure sensor	800
Data logger / controller	2,000
4G/LoRaWAN communication module	1,000
Solar panel + charge controller	1,000
Rechargeable battery	600
Weatherproof enclosure	500
Mounting mast / pole + brackets	1,000
Cables, connectors & lightning protection	600
Hardware subtotal	**15,500**
Site preparation & concrete foundation	1,500
Installation & sensor mounting	2,000
Installation subtotal	3,500
TOTAL	19,000

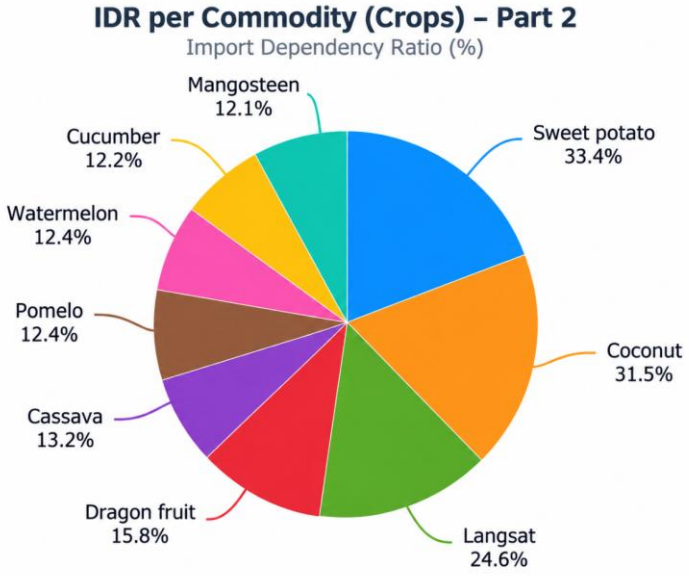
Appendix C2 – Budget & Potential Risk Estimation : Server & Application Development

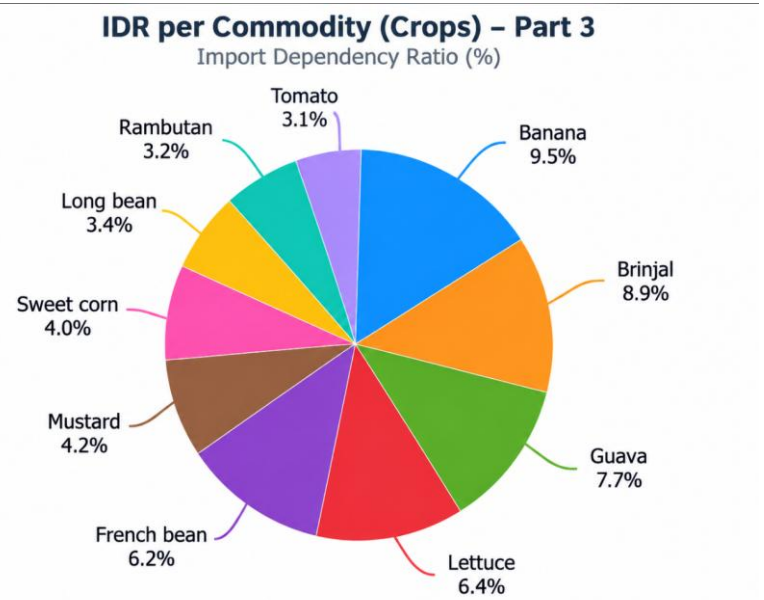
Component	Estimated cost
AWS IoT Core	RM80/month
Application/API server	RM150/month
Database	RM120/month
Cloud storage	RM50/month
Monitoring & security	RM50/month
Data transfer / miscellaneous	RM50/month
Estimated monthly total	**RM500/month**
Estimated annual total	**RM6,000/year**

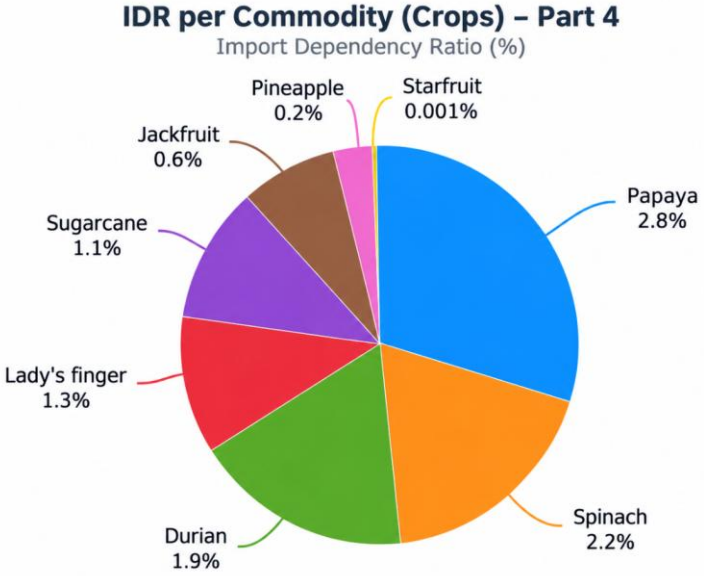
Development component	Estimated cost
UI/UX design	RM5,000
Web/mobile frontend	RM8,000
Backend & API development	RM8,000
Database & data integration	RM5,000
CRIS risk-analysis module	RM6,000
Scenario simulation & DSS	RM5,000
Authentication & security	RM2,000
Testing & deployment	RM1,000
Total	**RM40,000**

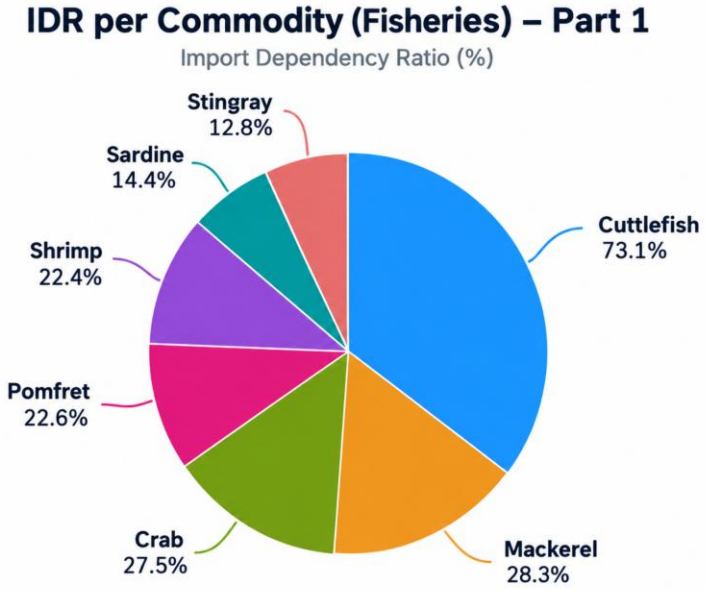
Appendix D: IDR & SSR

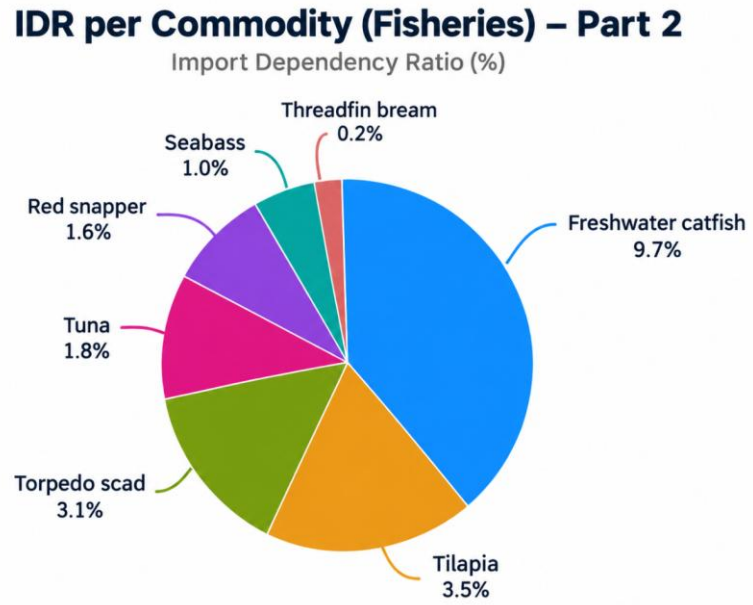






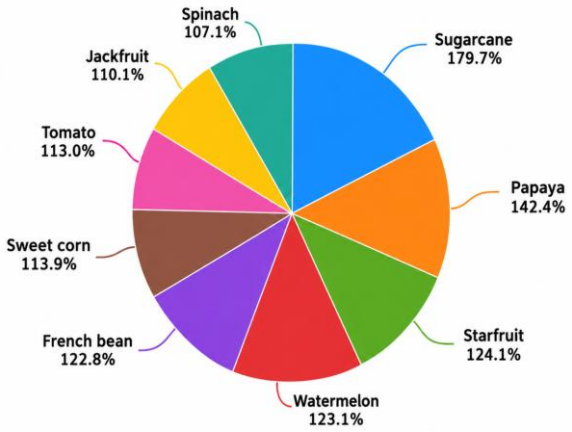






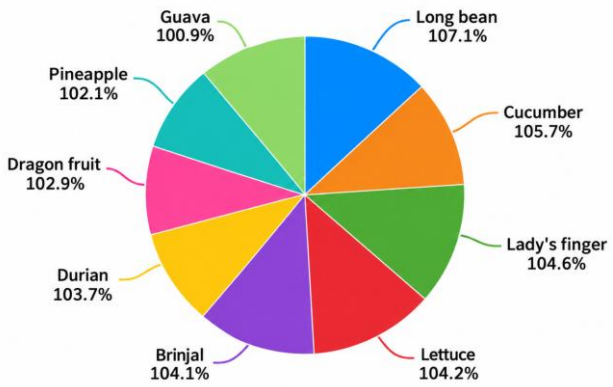
SSR per Commodity (Crops) – Part 1

Self-Sufficiency Ratio (%)



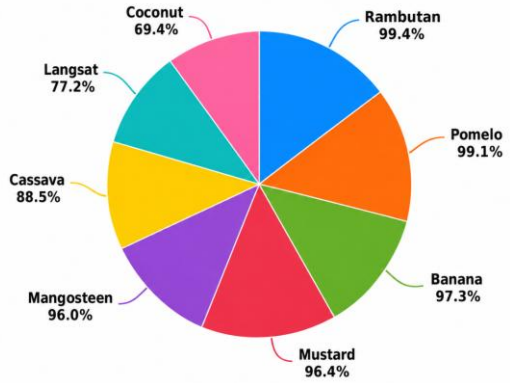
SSR per Commodity (Crops) – Part 2

Self-Sufficiency Ratio (%)



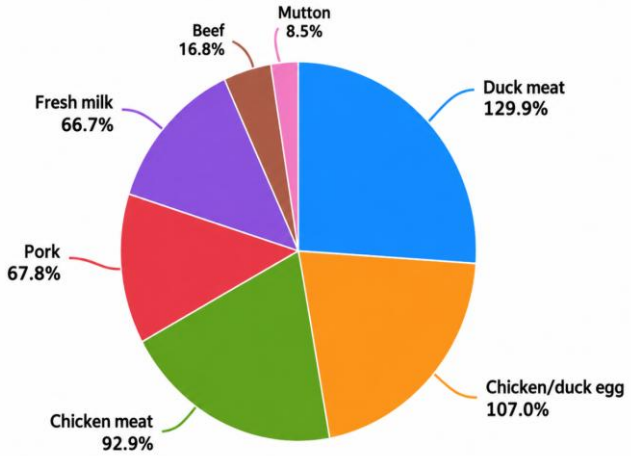
SSR per Commodity (Crops) – Part 3

Self-Sufficiency Ratio (%)



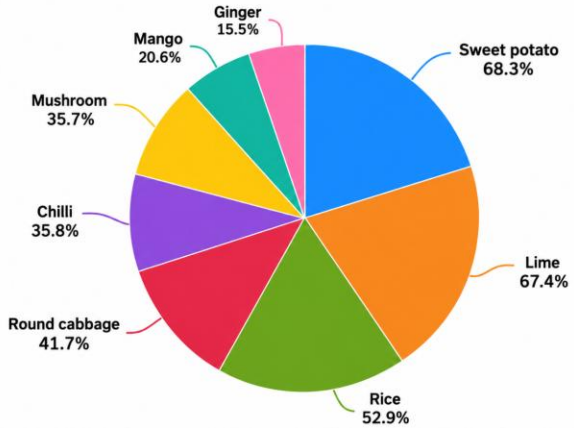
SSR per Commodity (Livestock)

Self-Sufficiency Ratio (%)



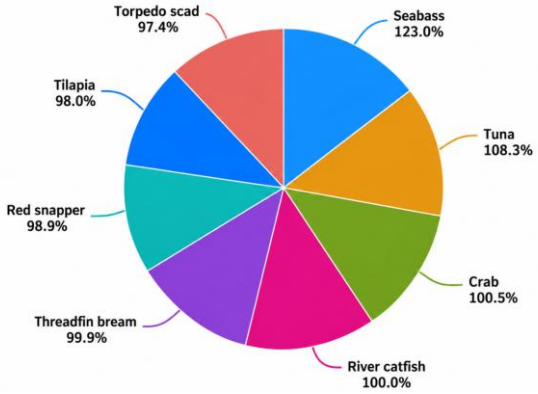
SSR per Commodity (Crops) – Part 4

Self-Sufficiency Ratio (%)



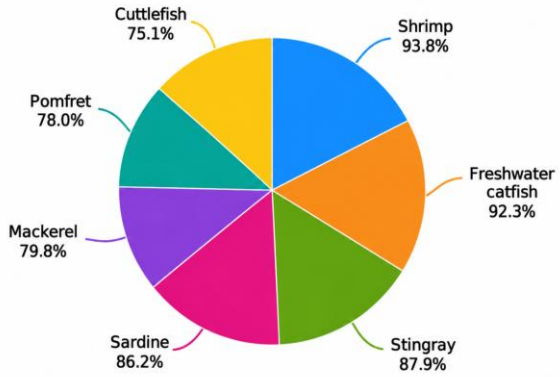
SSR per Commodity (Fisheries) – Part 1

Self-Sufficiency Ratio (%)



SSR per Commodity (Fisheries) – Part 2

Self-Sufficiency Ratio (%)



The ‘Blowout’ Of All Vulnerabilities At Once: What Happens If We Do Nothing?

PRODUCTION LOSS



2024 : Kedah experienced the paddy loss caused by flood:

- 57,323 tonnes of paddy
- 2,047 hectares of paddy fields
- RM158.7 million worth of paddy

The ‘Blowout’ Of All Vulnerabilities At Once: What Happens If We Do Nothing?

PRODUCTION LOSS



SUPPLY CHAIN DISCRUPTION



-DEC 2024: Output of 53000 tones of rice destroyed (BERNAMA)

“Disruption in local white-rice supply associated with floods and changing weather” ---
(Malaysia's Agriculture and Food Security Ministry)

The ‘Blowout’ Of All Vulnerabilities At Once: What Happens If We Do Nothing?

PRODUCTION LOSS



SUPPLY CHAIN DISCRUPTION



FOOD INSECURE ON VULNERABLE POPULATIONS



- Food prices increase
- Vulnerable group struggle to obtain sufficient food
- Malnutrition worsens and food insecurity happen on vulnerable group :
  A 10% increase in food prices is associated with:
  +3.5% moderate/severe food insecurity
  +2.7–4.3% child wasting
  +4.8–6.1% severe child wasting

---FAO, IFAD, UNICEF, WFP & WHO, *SOFI 2025*

The ‘Blowout’ Of All Vulnerabilities At Once: What Happens If We Do Nothing?

PRODUCTION LOSS



SUPPLY CHAIN DISCRUPTION



FOOD INSECURE ON VULNERABLE POPULATIONS



POVERTY AND INEQUALITY INCREASE



Droughts have affected more than 66 million people in Southeast Asia over the past 30 years.

≈ 80% OF DROUGHT'S ECONOMIC IMPACT

is absorbed by agriculture

ASEAN, Ready for the Dry Years (2020)

The ‘Blowout’ Of All Vulnerabilities At Once: What Happens If We Do Nothing?

PRODUCTION LOSS



SUPPLY CHAIN DISCRUPTION



FOOD INSECURE ON VULNERABLE POPULATIONS



POVERTY AND INEQUALITY INCREASE



Recovery becomes more expensive and difficult

The World Bank estimates that a hypothetical **1-in-20-year flood could cost Malaysia up to 4.1% of GDP in 2030**.